Chapter 1 · Part 1

Computer Networks: The Big Picture

How applications communicate across the Internet

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CSE589 · Modern Network Concepts · Fall 2026

Today: build a first mental model of the Internet

By the end of this lesson, you should be able to…

1. **Internet service** — Explain how applications communicate over the Internet.
2. … Packet path
3. … Network structure
4. … Protocol layers

Begin with the Internet service used by applications.

Today: build a first mental model of the Internet

By the end of this lesson, you should be able to…

1 ✓

Internet service

Explain how applications communicate over the Internet.

2

Packet path

Trace a packet through access networks and the core.

3

…

Network structure

4

…

Protocol layers

Trace one packet through an access network and the Internet core.

Today: build a first mental model of the Internet

By the end of this lesson, you should be able to…

1 ✓
Internet service
Explain how applications communicate over the Internet.

2 ✓
Packet path
Trace a packet through access networks and the core.

3
Network structure
Distinguish edge devices, access networks, and the core.

4
...
Protocol layers

Identify the network edge, access networks, and the core.

Today: build a first mental model of the Internet

By the end of this lesson, you should be able to…

1 ✓

Internet service

Explain how applications communicate over the Internet.

2 ✓

Packet path

Trace a packet through access networks and the core.

3 ✓

Network structure

Distinguish edge devices, access networks, and the core.

4

Protocol layers

Match segments, packets, frames, and signals to layers.

Connect segments, packets, frames, and signals to their layers.

Networks let applications communicate beyond one machine

One network supports many kinds of application activity.

1 **Communicate**

Messages and conversations reach people on other devices.

Next

2 **Move data**

Next

3 **Remote services**

Networks let people communicate across devices and distance.

Networks let applications communicate beyond one machine

One network supports many kinds of application activity.

✓ **Communicate**

Messages and conversations reach people on other devices.

2 **Move data**

Files, requests, and results travel between applications.

Next

3 **Remote services**

Networks move application data beyond one machine.

Networks let applications communicate beyond one machine

One network supports many kinds of application activity.

✓ **Communicate**

Messages and conversations reach people on other devices.

✓ **Move data**

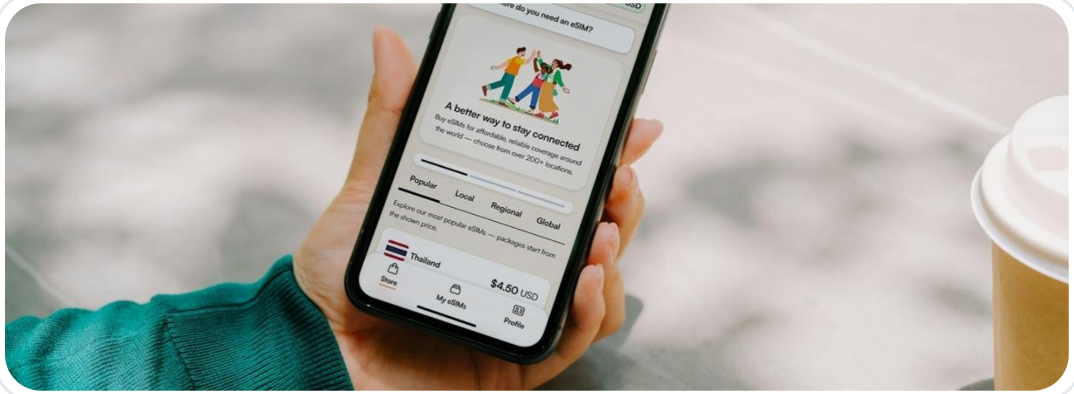
Files, requests, and results travel between applications.

3 **Remote services**

Streaming, cloud tools, and computation run beyond one machine.

Networks make remote services feel local and immediate.

The same Internet looks simple outside and complex inside



User / application view

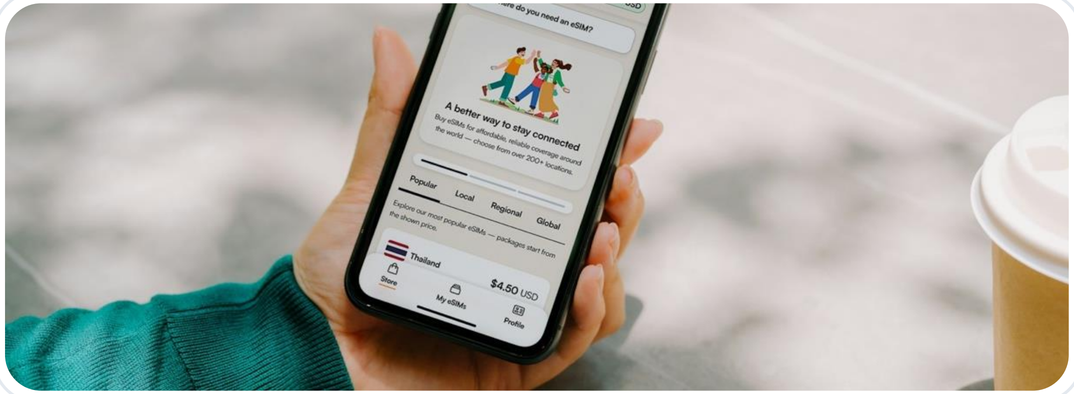
A simple communication service: send data, receive data, reach another application.

Then look inside

Network / system view

From the outside, the Internet feels like a simple service.

The same Internet looks simple outside and complex inside



User / application view

A simple communication service: send data, receive data, reach another application.

Network / system view

Hosts, access networks, routers, links, and protocols cooperate behind the service.

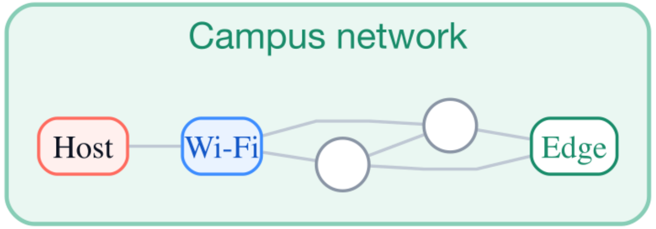
Inside, many systems cooperate to provide that simple service.

Applications use the Internet as a communication service

The application asks for communication; it does not choose every router and link.

Inside, the Internet is a network of interconnected networks

Start with the sender's campus network.



A local network contains devices and links managed as one system.

Inside, the Internet is a network of interconnected networks

The campus network connects to an independently operated ISP.

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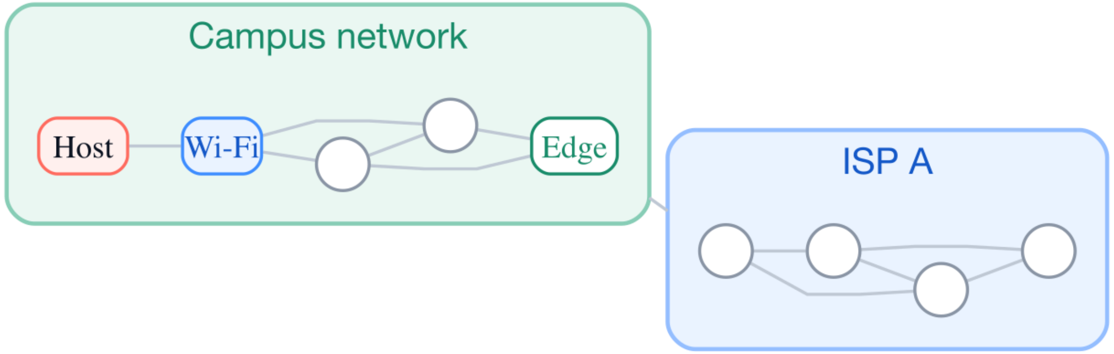
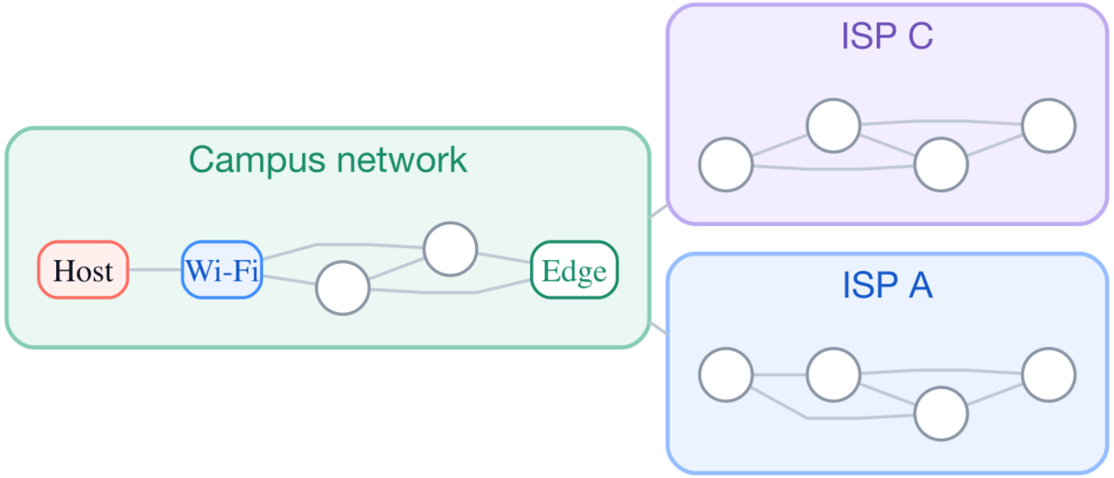


Connecting another network extends the path beyond the campus.

Inside, the Internet is a network of interconnected networks

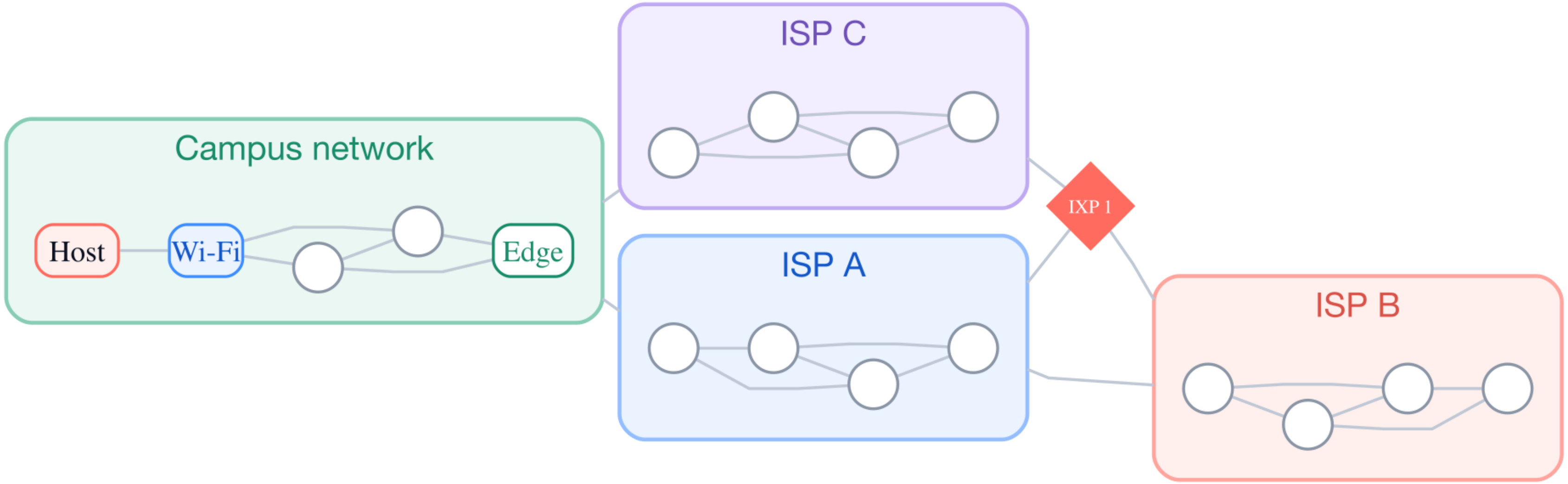
A second ISP creates another way out of the campus network.



Two providers create alternative exits from the same campus network.

Inside, the Internet is a network of interconnected networks

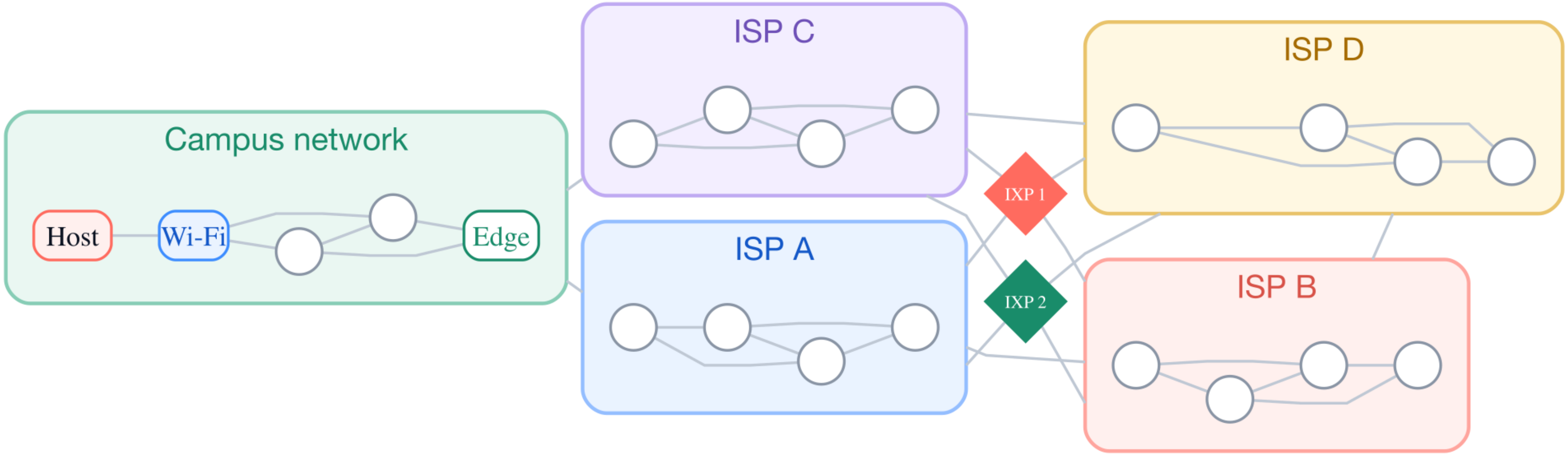
Another ISP joins through direct links and an exchange point.



Networks can interconnect directly or through an exchange point.

Inside, the Internet is a network of interconnected networks

More providers create still more path choices.



Each added provider creates new paths through the system.

Inside, the Internet is a network of interconnected networks

Several independently operated networks connect through multiple paths.

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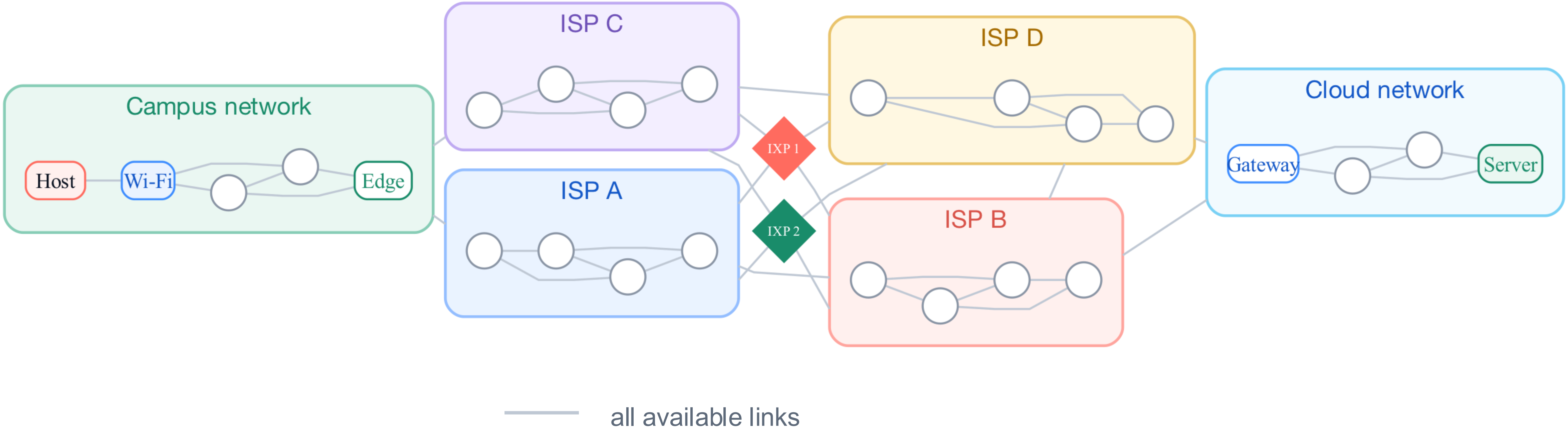
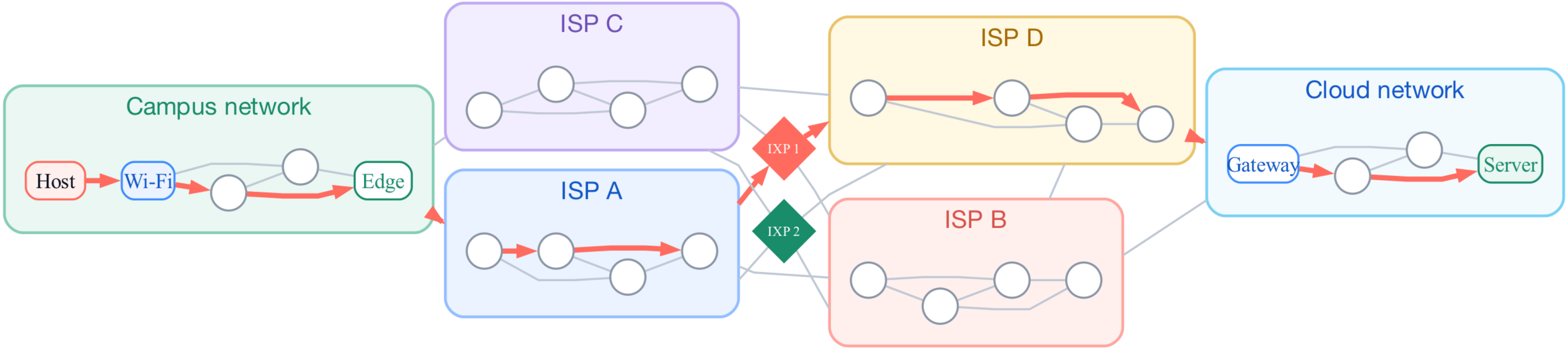


The Internet is interconnected networks—not one network controlled by one operator.

A packet follows one possible route through the topology

The topology offers alternatives; this packet uses only one of them.



other available links | one possible route

A route selects a sequence of links; it does not use the whole topology.

How does a complex system provide a simple service?

What the application sees

A destination

Data to send

Data or a result returned

A relatively small communication interface

How does a complex system provide a simple service?

2 / 3

What the application sees

A destination

Data to send

Data or a result returned

A relatively small communication interface

What the network must do

Prepare packets

Reach the first router

Forward across many networks

Transmit over multiple physical links

How does a complex system provide a simple service?

What the application sees

A destination

Data to send

Data or a result returned

A relatively small communication interface

What the network must do

Prepare packets

Reach the first router

Forward across many networks

Transmit over multiple physical links

To connect the two views, follow one communication from beginning to end.

Follow one communication

Use one application request as a running example

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10 MB assignment

1 **Sender** A laptop connected to campus Wi-Fi

Use one application request as a running example

10 MB assignment

1. **Sender** — A laptop connected to campus Wi-Fi
2. **Destination** — A remote server that stores the submission

Use one application request as a running example

10 MB assignment

1. **Sender** — A laptop connected to campus Wi-Fi
2. **Destination** — A remote server that stores the submission
3. **Question** — What must happen between click and completion?

Use one application request as a running example

10 MB assignment

1. **Sender** — A laptop connected to campus Wi-Fi
2. **Destination** — A remote server that stores the submission
3. **Question** — What must happen between click and completion?

We will follow the same packet through every device and every link.

Laptop → Wi-Fi AP → Campus → Edge router → Internet core → Server

The endpoints see a service—not the internal hops

The upload begins at one endpoint and finishes at another.

Sender
Sending side
Upload application

→

A black box to the endpoints

● → ● → ● → ● → ●

Many internal hops

access networks · routers · links · other networks

→

Receiver
Receiving side
Storage application

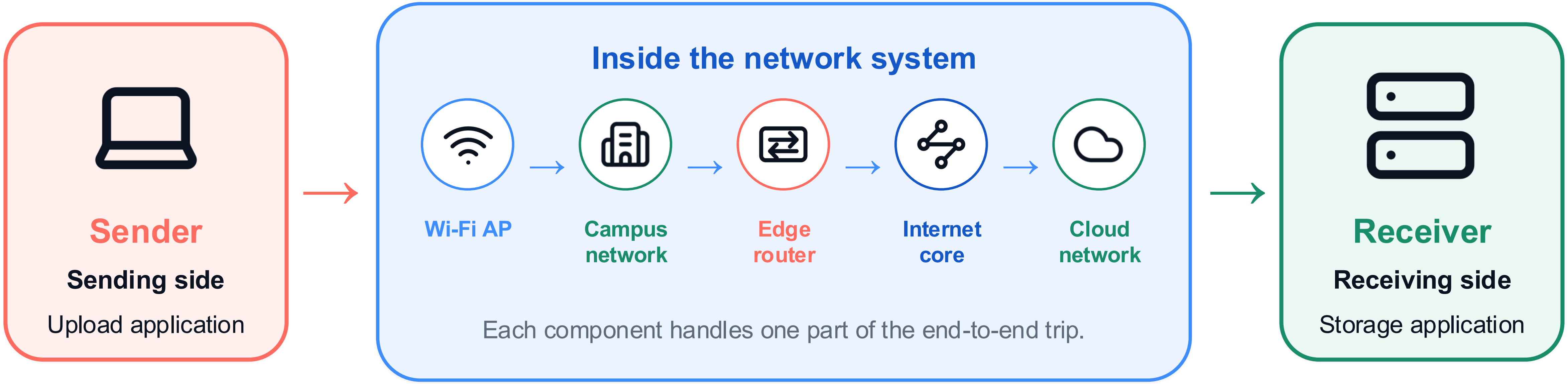
The service is visible at the endpoints; the internal route is not.

Laptop → Wi-Fi AP → Campus → Edge router → Internet core → Server

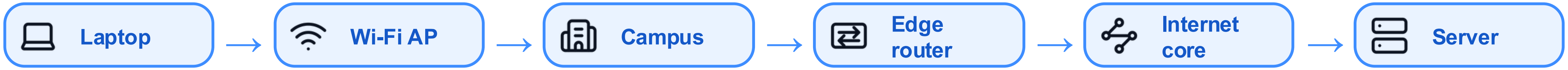
Follow one communication

Opening the black box reveals the network path

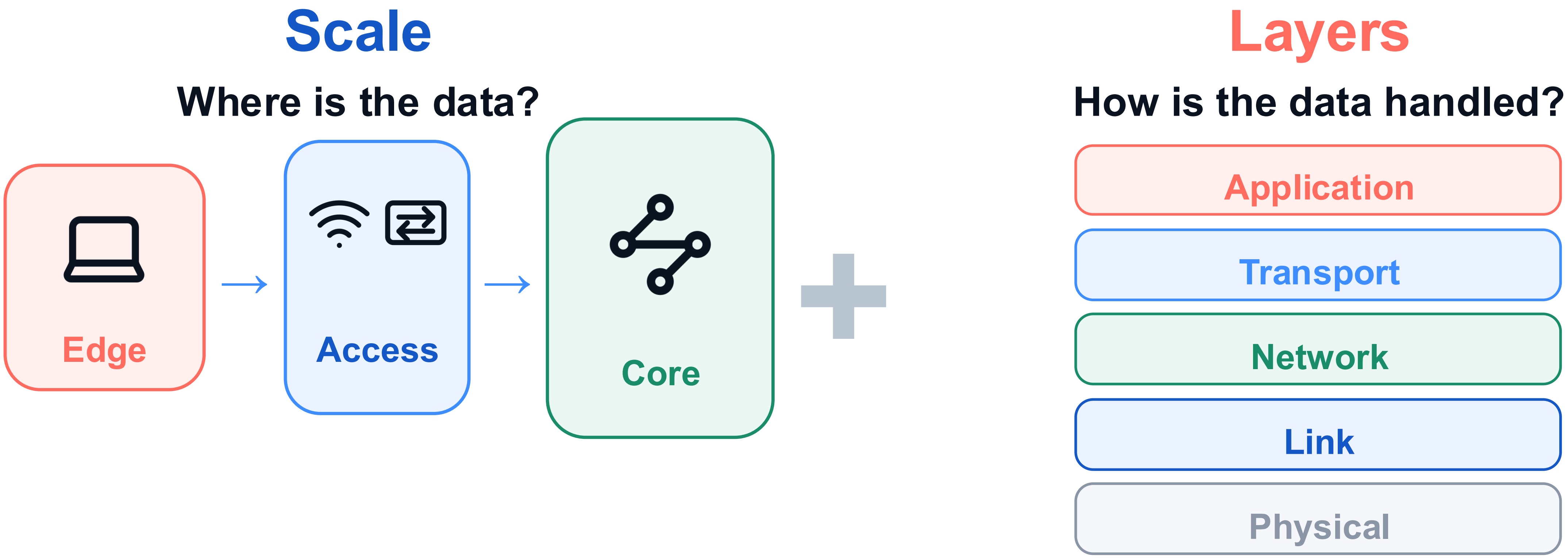
Now the systems between the two endpoints become visible.



Opening the black box reveals the systems that implement the service.

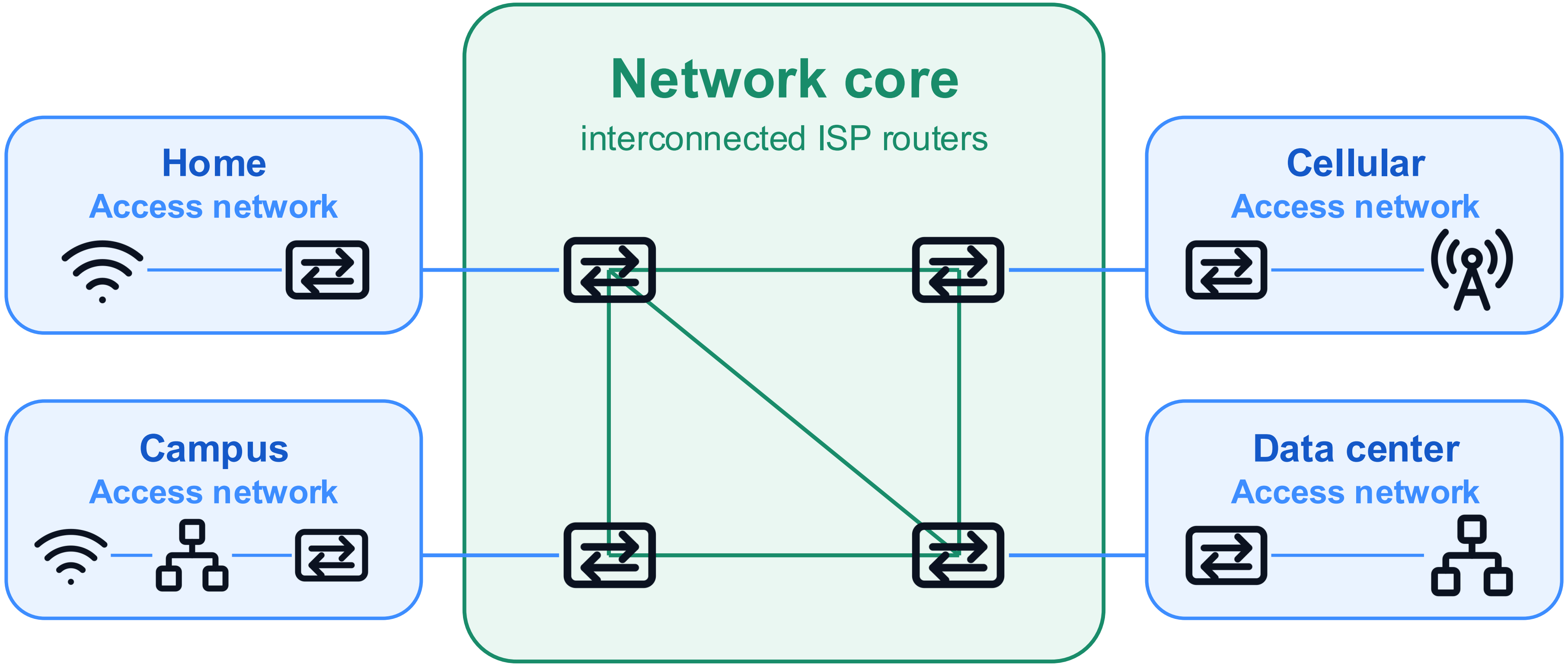


Two views organize the rest of the journey



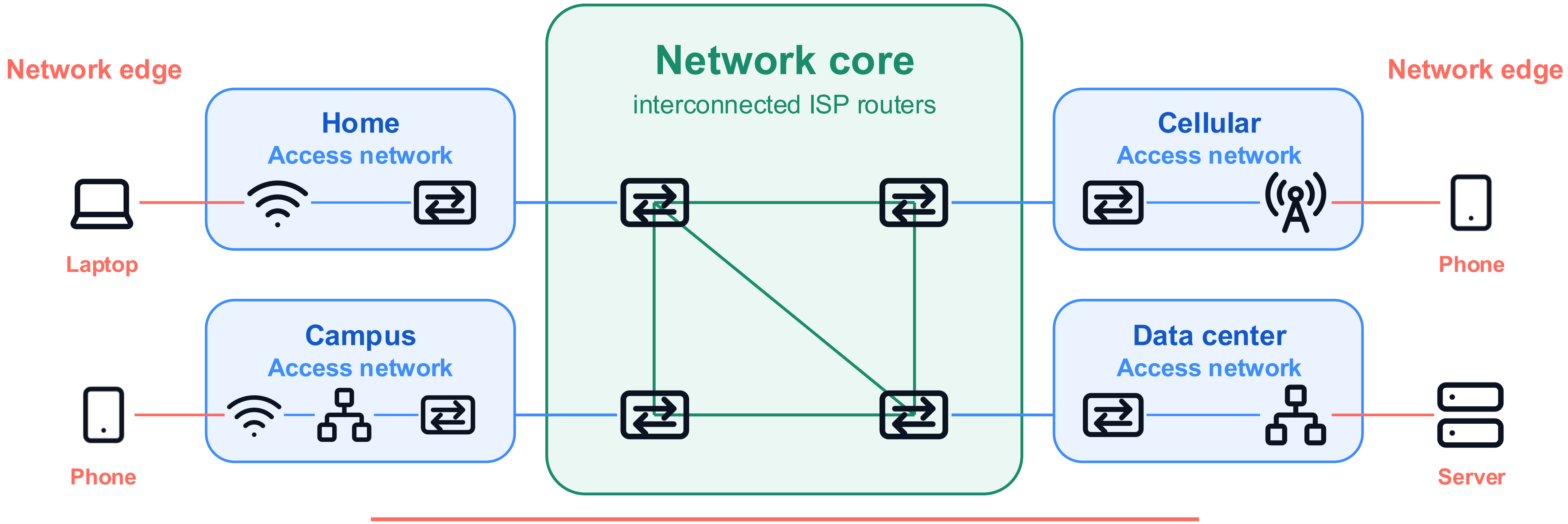
Two views of the same communication

The core connects many access networks



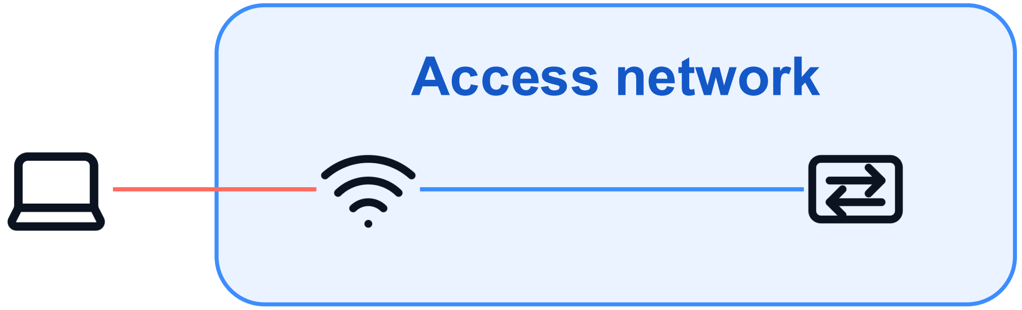
Access networks attach to the core at their first routers.

Edge devices reach the core through an access network

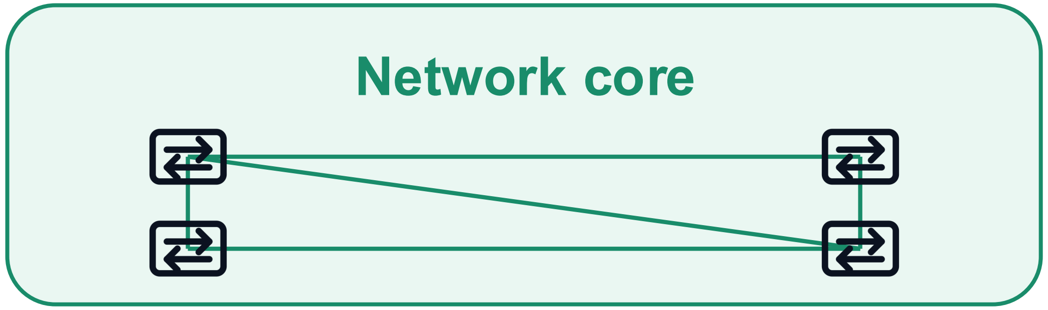


Edge devices use an access network to reach the core.

Access and core solve different connection problems



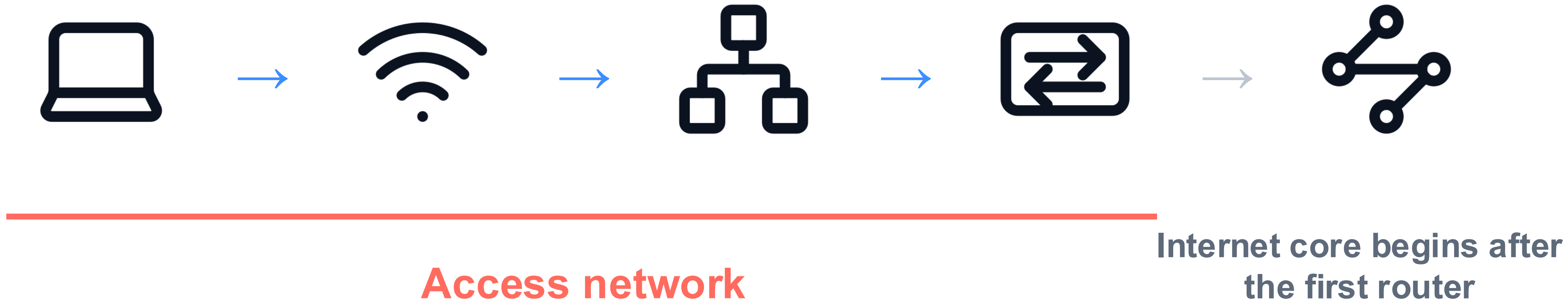
- Begins at an endpoint
- Uses Wi-Fi, cellular, or wired links
- Ends at the first router



- Begins after the first router
- Crosses routers and provider networks
- Connects many access networks

Access gets a device into the network; the core carries packets between networks.

The access network reaches the first router



Definition: the access network connects an end system to its first router.

A Wi-Fi access point is not the first router

Access point

Connects wireless devices to a local network

Receives and transmits Wi-Fi frames

Usually does not decide the Internet-wide route

Router

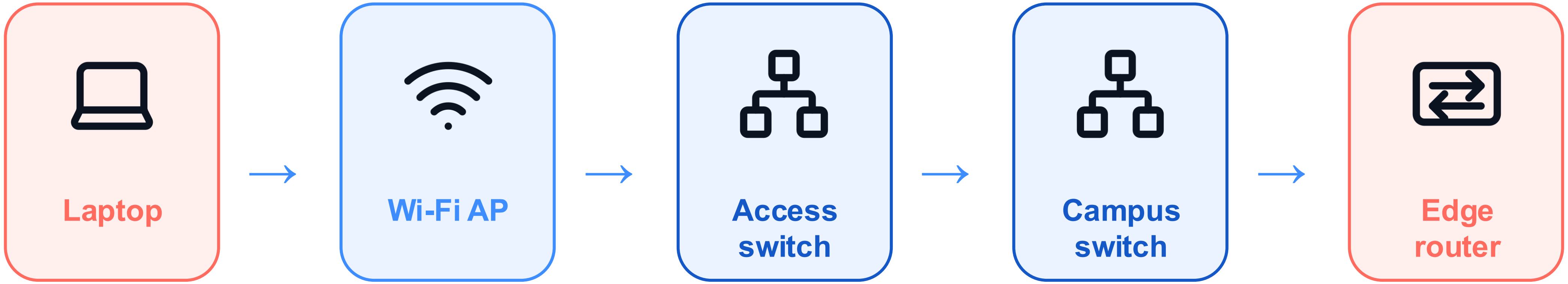
Connects different networks

Examines packet delivery information

Chooses an outgoing link toward the destination

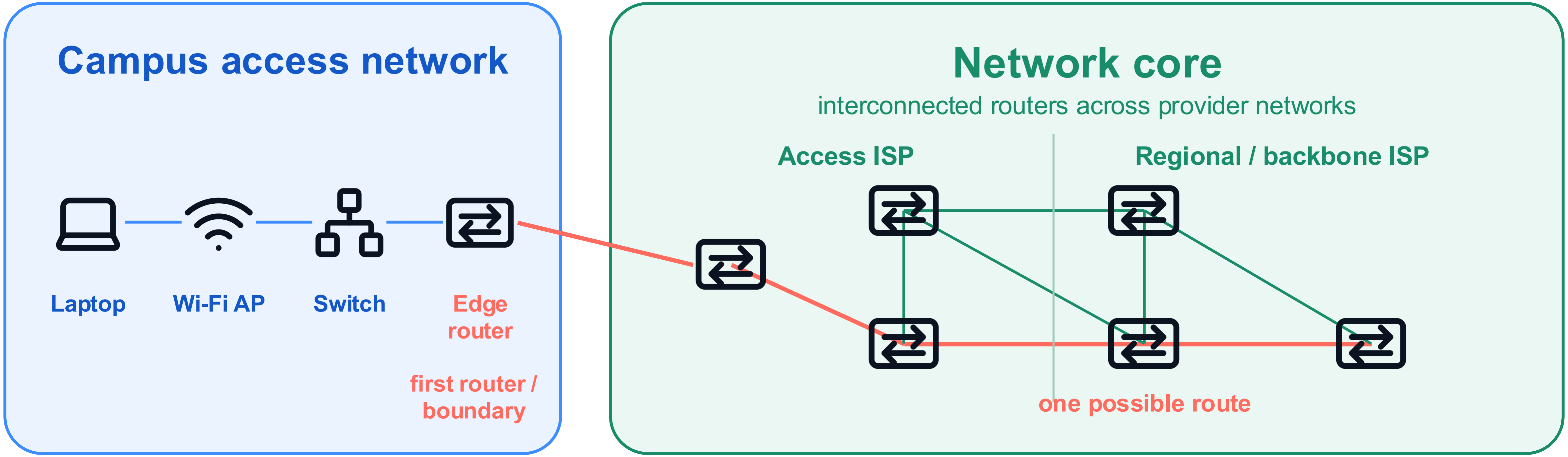
A home "Wi-Fi router" often combines both roles in one box; the roles are still different.

A campus access network contains many local links



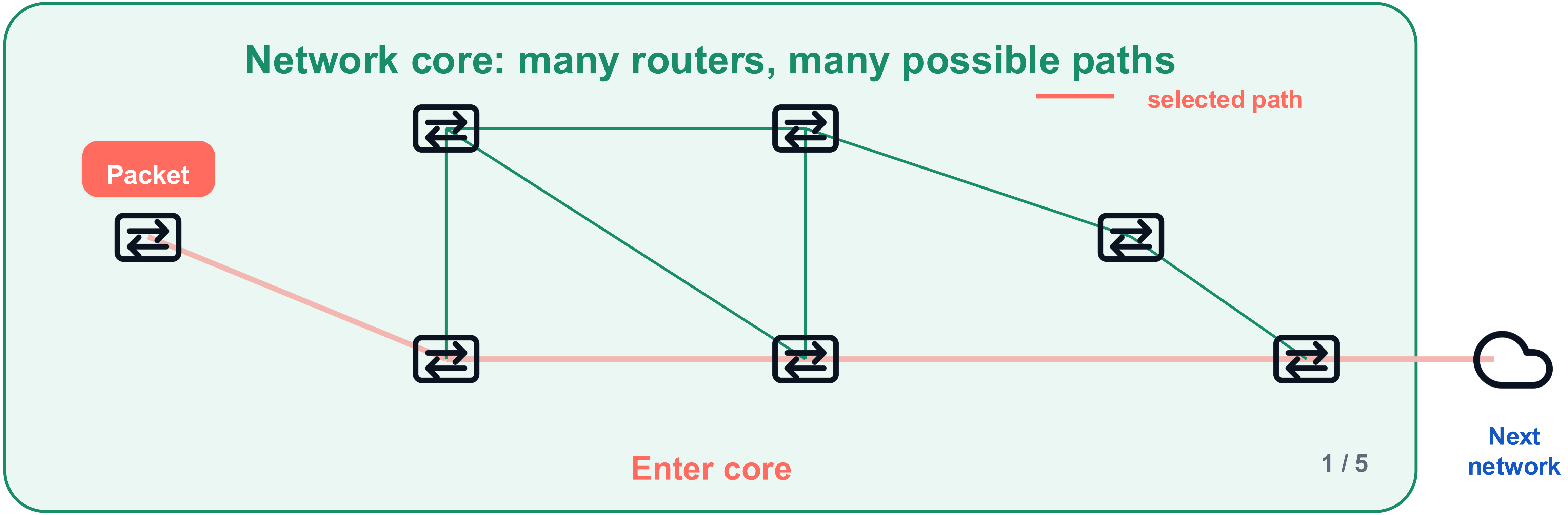
One access network can include several local hops before the first edge router.

The edge router connects the access network to the core



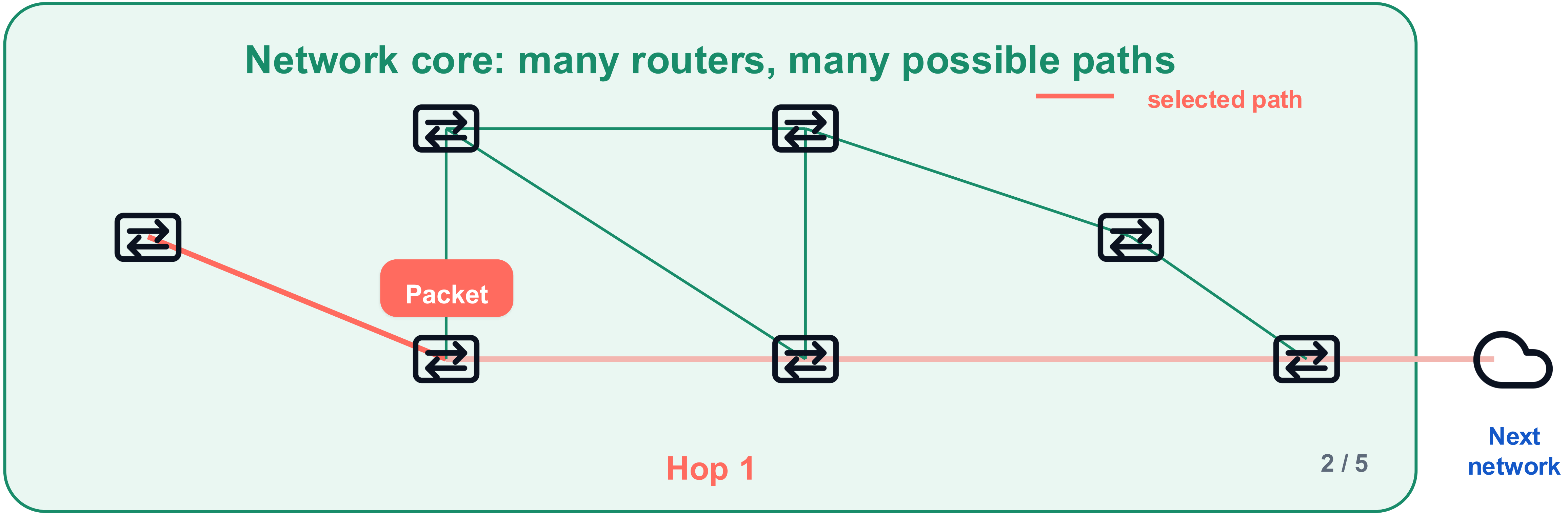
The access network ends at the first router; the core continues across many provider routers.

A packet moves through the core one router at a time



The packet crosses the core one next-hop transmission at a time.

A packet moves through the core one router at a time



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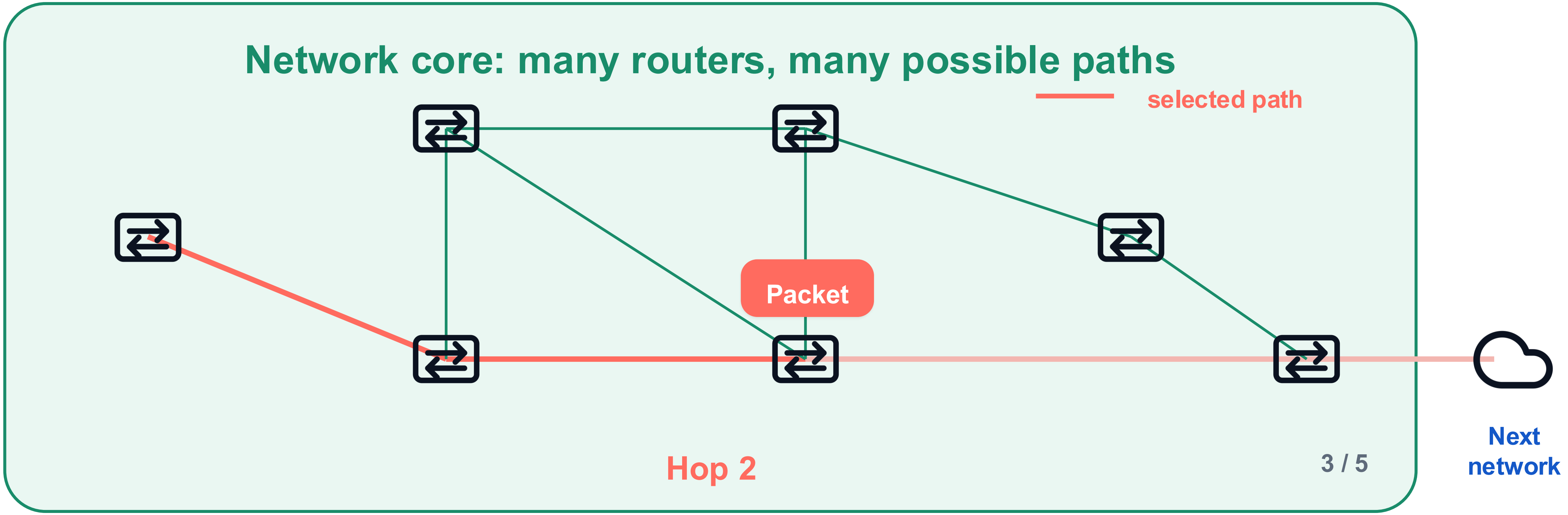

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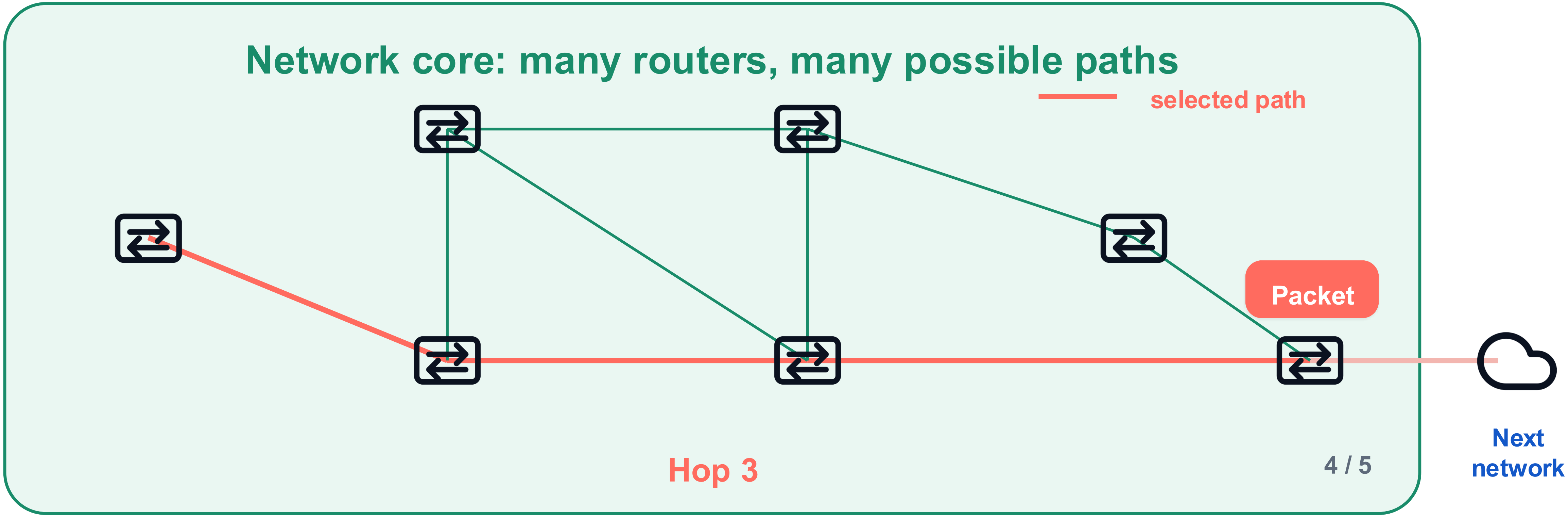

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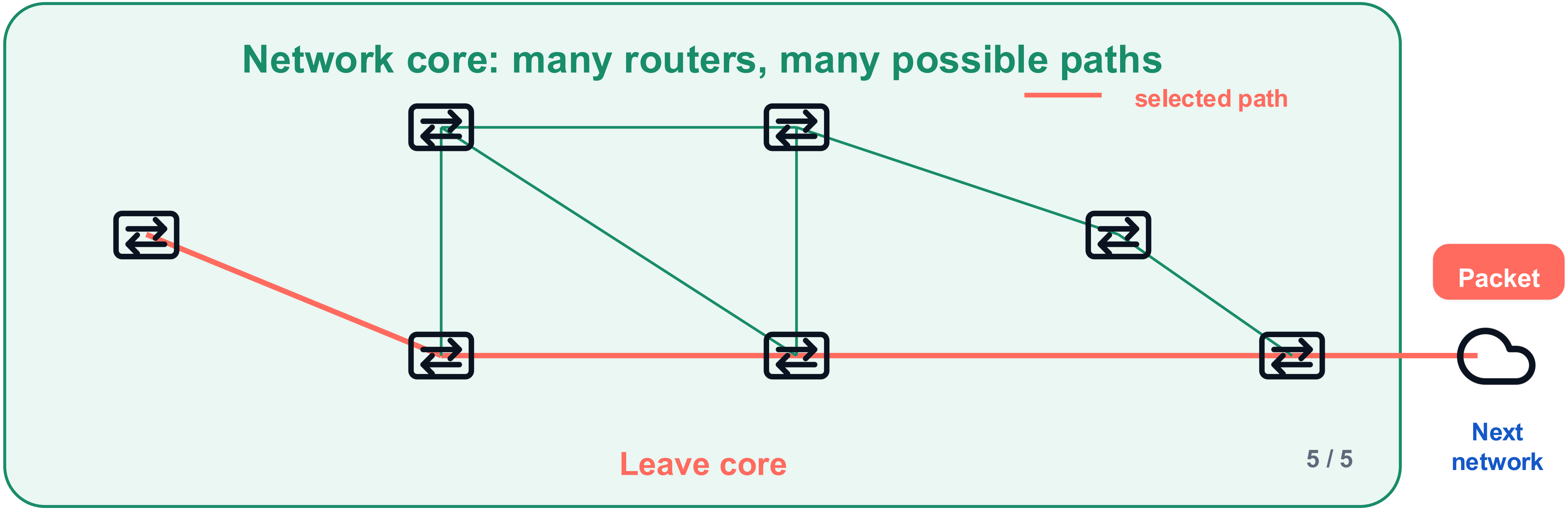
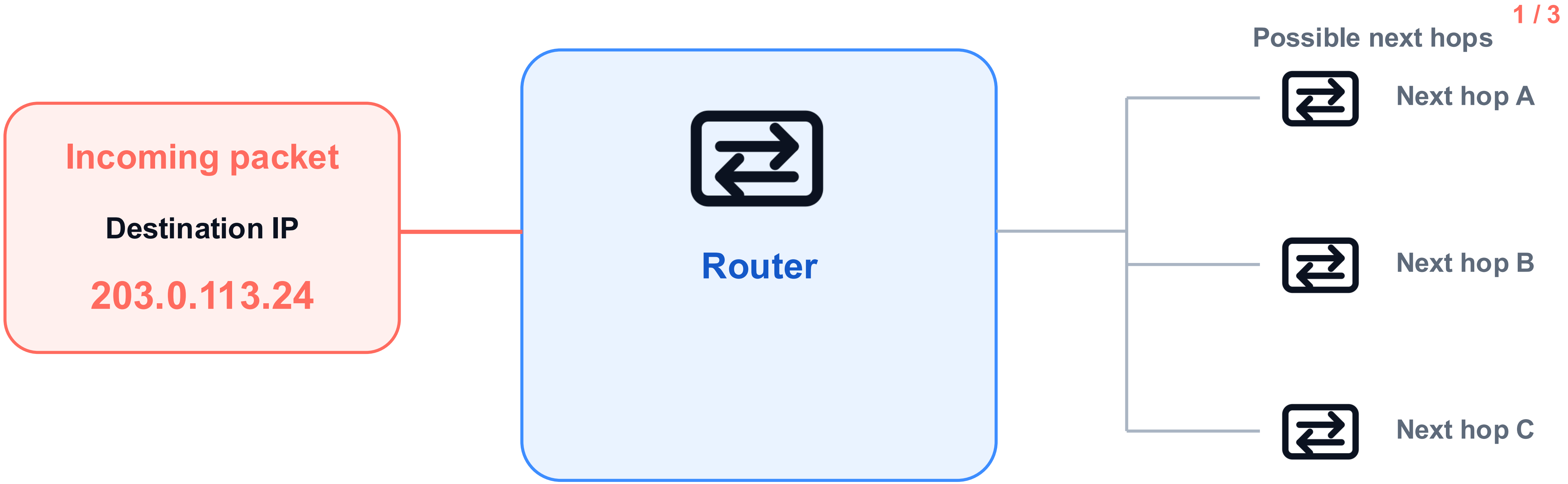


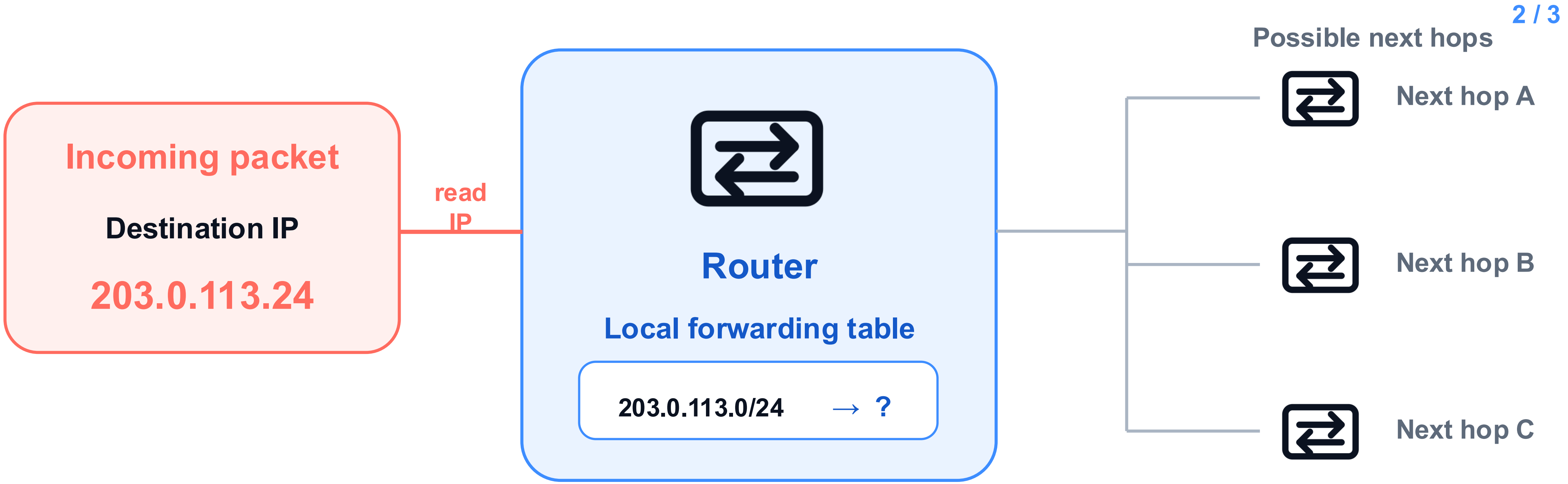
The packet crosses the core one next-hop transmission at a time.

A router receives, analyzes, and forwards one packet



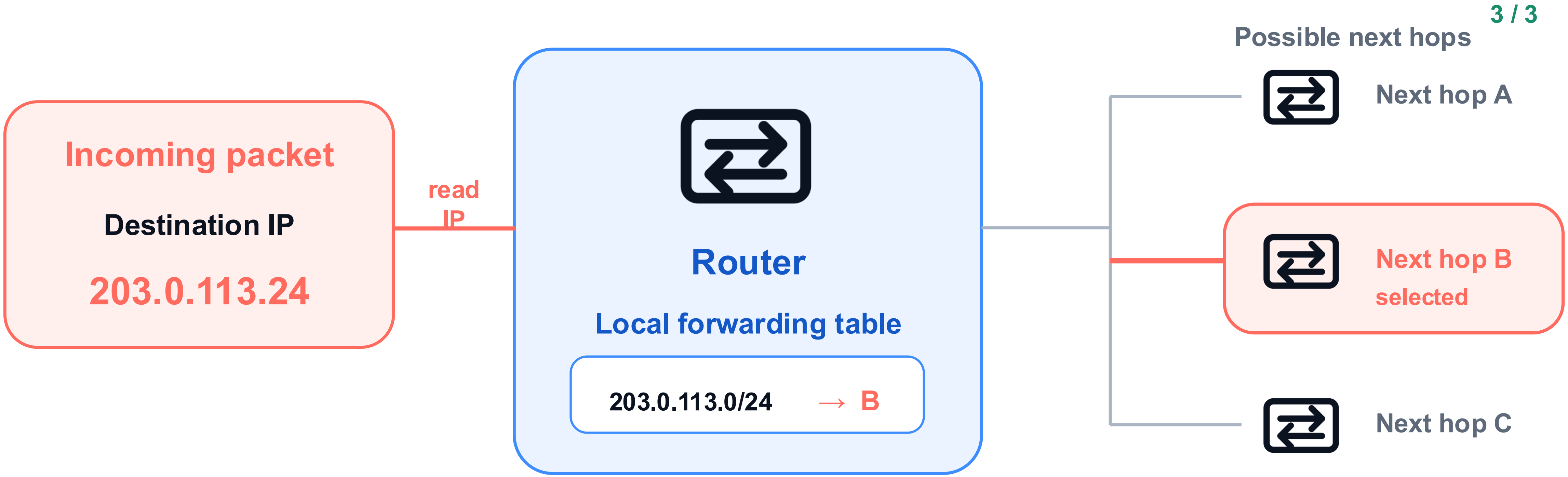
1 · Receive: a packet arrives on one input interface.

A router receives, analyzes, and forwards one packet



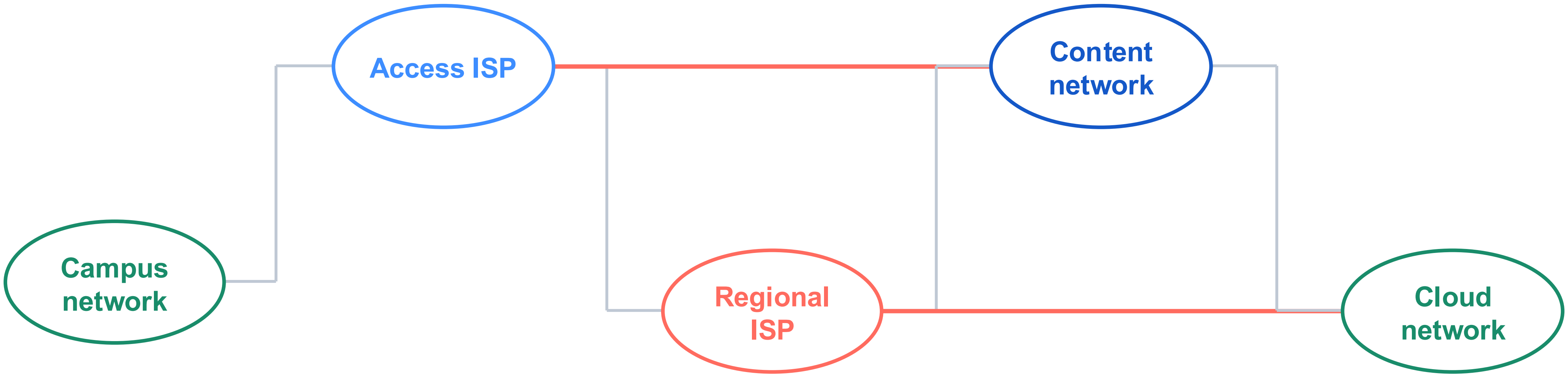
2 · Analyze: read the destination IP and match the local forwarding table.

A router receives, analyzes, and forwards one packet



3 · Forward: select one outgoing interface and one next-hop router.

The core is not one network owned by one organization



Each bubble is independently operated. The Internet emerges when these networks interconnect.

Networks interconnect in several ways

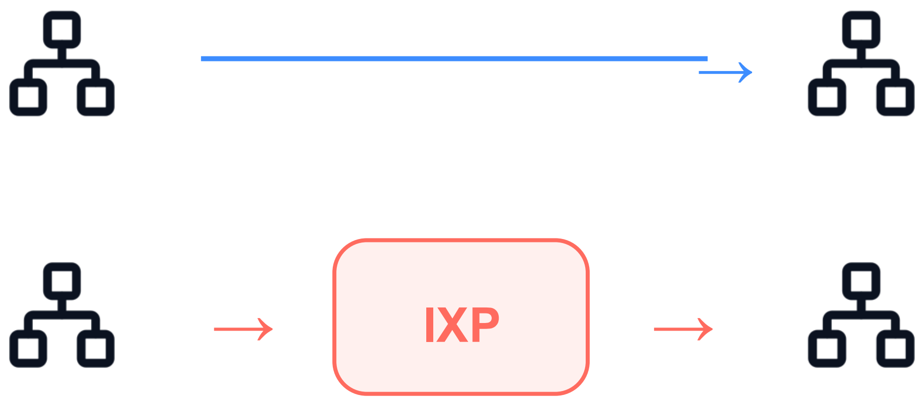
1 / 3

Direct connection

Two networks connect to each other directly.

Direct connections join two networks without an intermediary.

Networks interconnect in several ways



Direct connection

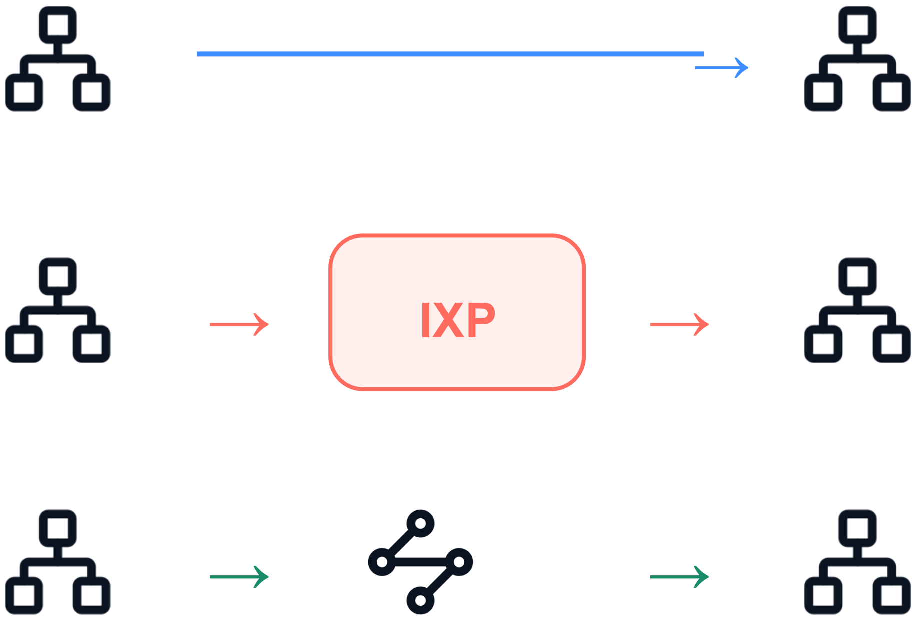
Two networks connect to each other directly.

Internet exchange

Several networks meet at a shared exchange point.

An Internet exchange lets several networks meet at one shared location.

Networks interconnect in several ways



Direct connection

Two networks connect to each other directly.

Internet exchange

Several networks meet at a shared exchange point.

Provider path

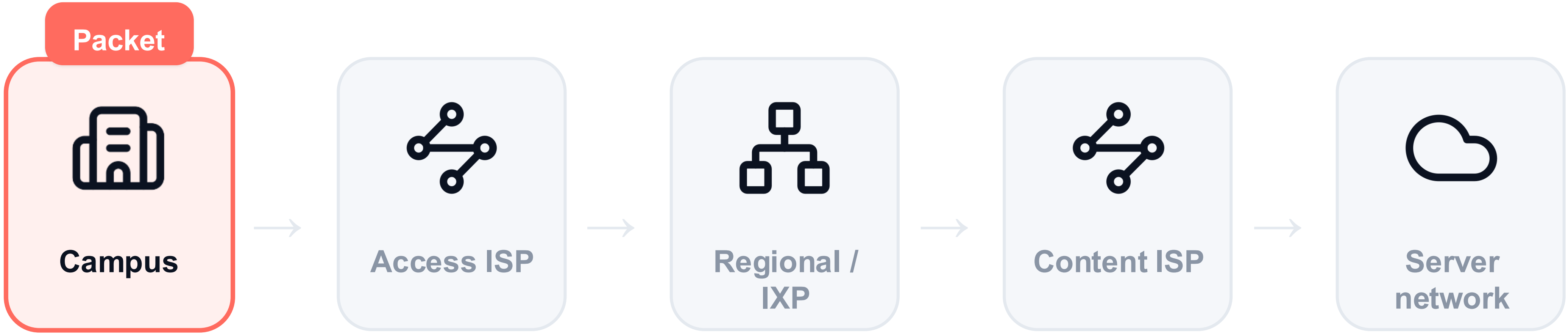
One network pays another to reach the wider Internet.

Provider relationships extend reach—and real routes often combine all three patterns.

View 1 · Scale

One packet crosses one network boundary at a time

1 / 5

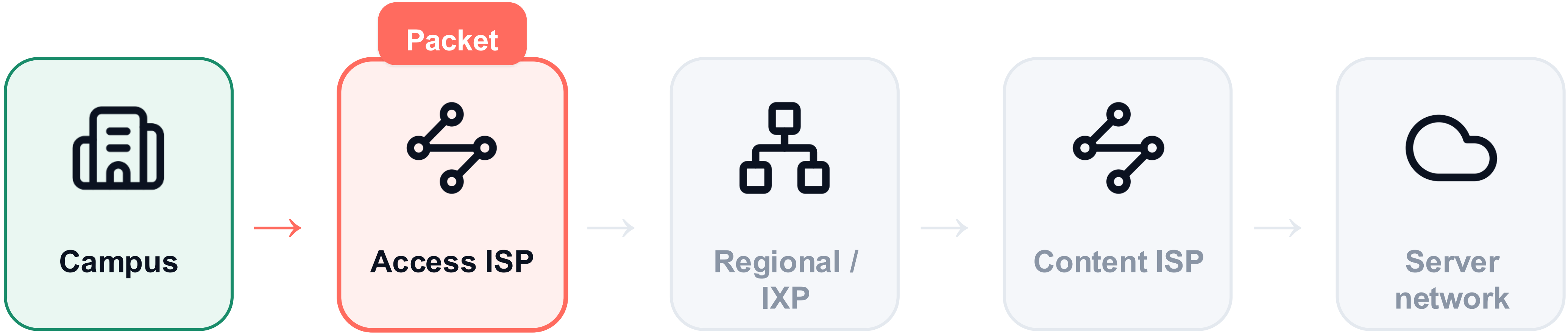


Now inside: Campus

The packet advances across one administrative network boundary at a time.

One packet crosses one network boundary at a time

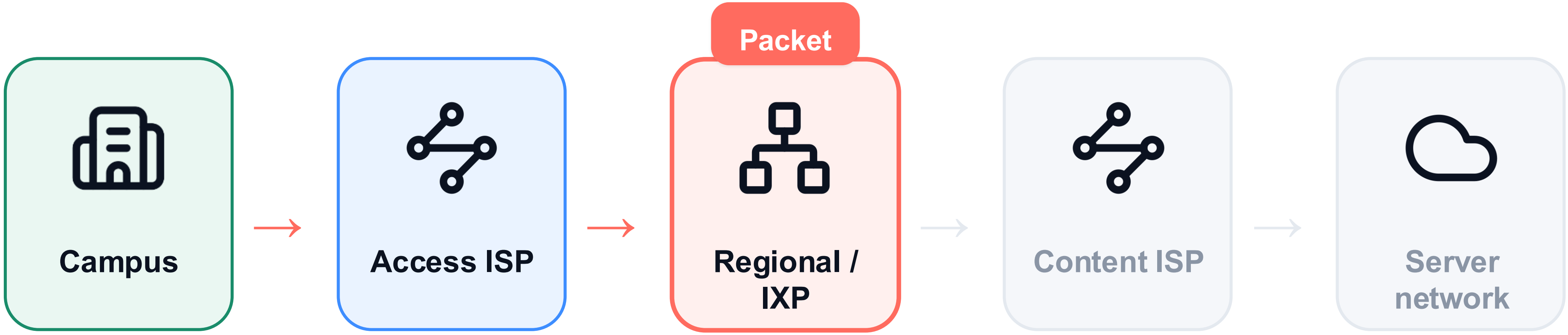
2 / 5



Now inside: Access ISP

The packet advances across one administrative network boundary at a time.

One packet crosses one network boundary at a time



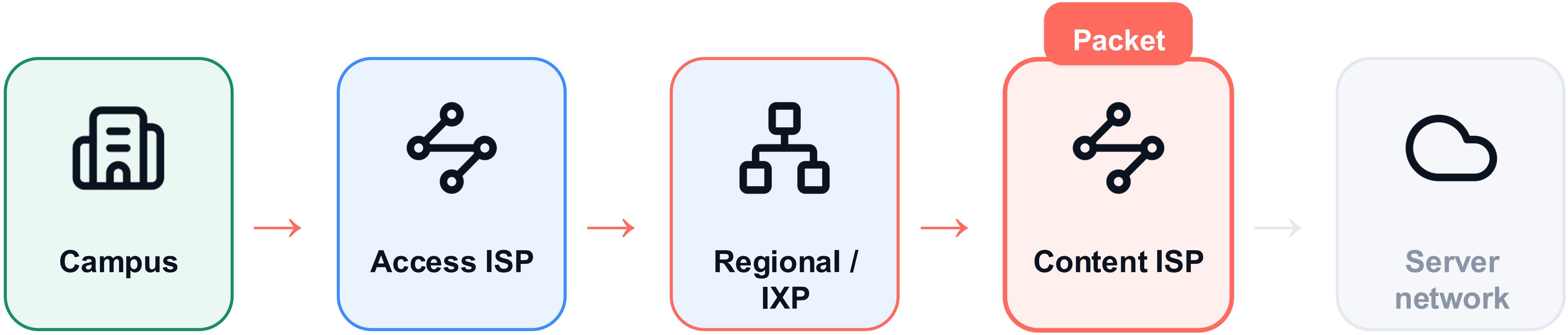
Now inside: Regional / IXP

The packet advances across one administrative network boundary at a time.

View 1 · Scale

One packet crosses one network boundary at a time

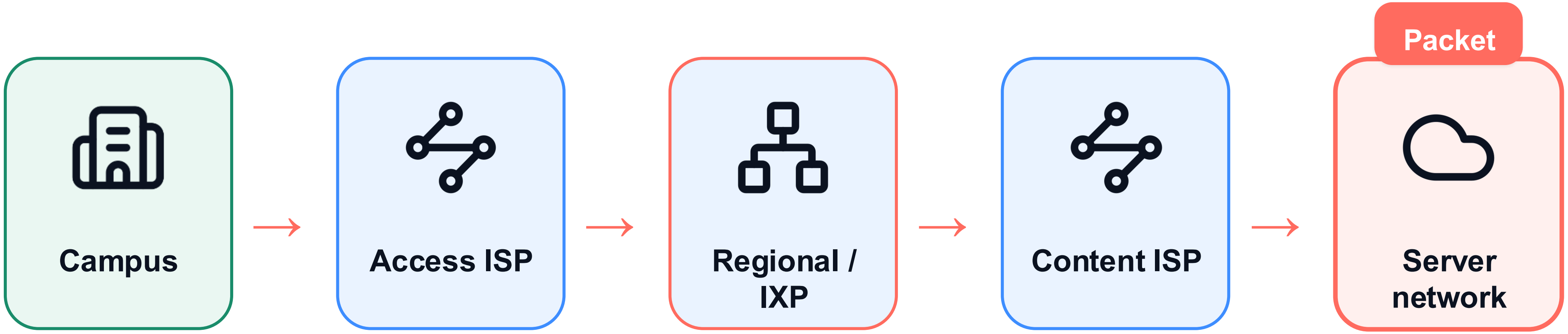
4 / 5



Now inside: Content ISP

The packet advances across one administrative network boundary at a time.

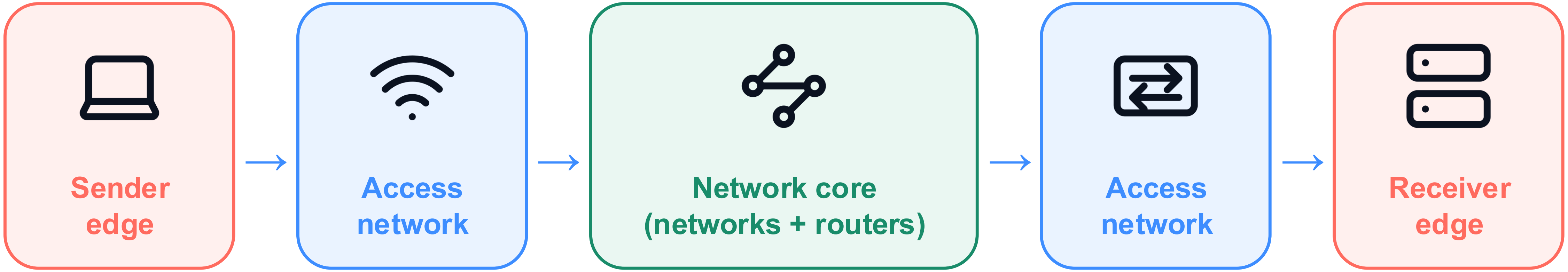
One packet crosses one network boundary at a time



Now inside: Server network

The packet advances across one administrative network boundary at a time.

Local access leads into a network of networks



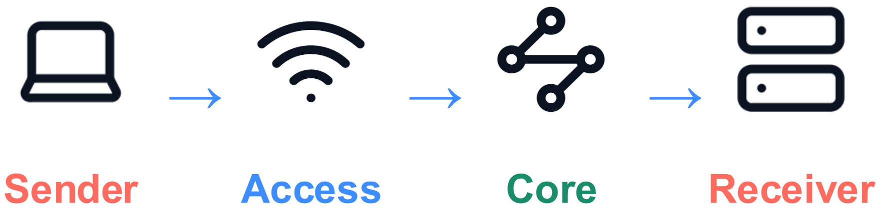
endpoint → local access → interconnected core → local access → endpoint

This is the geographic and organizational path. Next we inspect what each host and link does to the data.

Scale follows the path; layers explain the work

Scale view

Where is the data traveling?



One journey across devices and networks

Layer view

What work is being done to the data?

Application	define meaning
Transport	deliver between apps
Network	route across networks
Link	move one hop
Physical	send signals

Scale tracks movement across the network; layers track processing inside each host.

The application begins with a file and its meaning

assignment.pdf

→

Application meaning

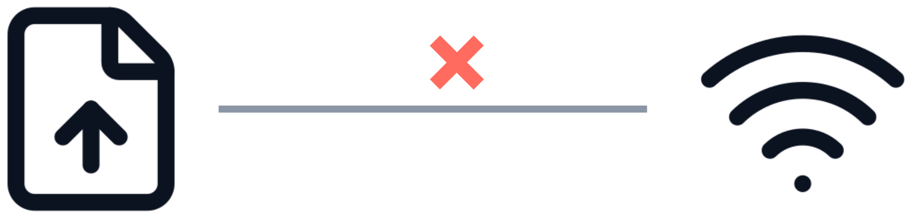
“Upload this file to the course server.”

The application understands the operation and expected response. The layers below carry bytes without understanding the assignment.

Top-down reasoning begins with what the application wants to accomplish.

A file does not jump directly to Wi-Fi

Not one direct jump



File — Wi-Fi

The real path inside the sender

Application	meaning
Transport	segments
Network	packets
Link	frames
Physical	signals

↓ one layer at a time

The application hands bytes downward; each layer prepares them for the layer below.

Application bytes move down to transport

Application	**meaning**
Transport	segments
Network	packets
Link	frames
Physical	signals

Application layer

application bytes

↓

handoff to transport

The application defines meaning; transport receives a sequence of bytes.

Transport divides data and adds a header to every segment

Application	meaning
Transport	**segments**
Network	packets
Link	frames
Physical	signals

1 / 3

Transport layer

application byte stream

1 · Transport receives the application's byte stream.

Transport divides data and adds a header to every segment

- **Application** — meaning
- **Transport** — segments
- **Network** — packets
- **Link** — frames
- **Physical** — signals

2 / 3

Transport layer

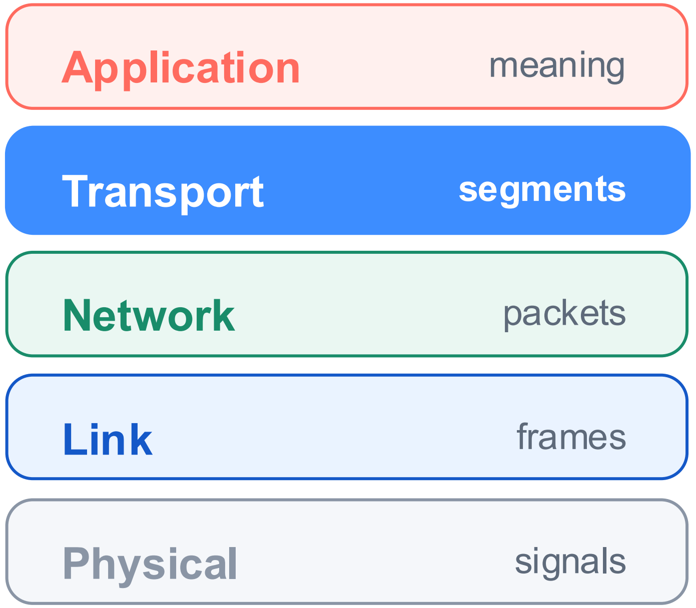
application byte stream

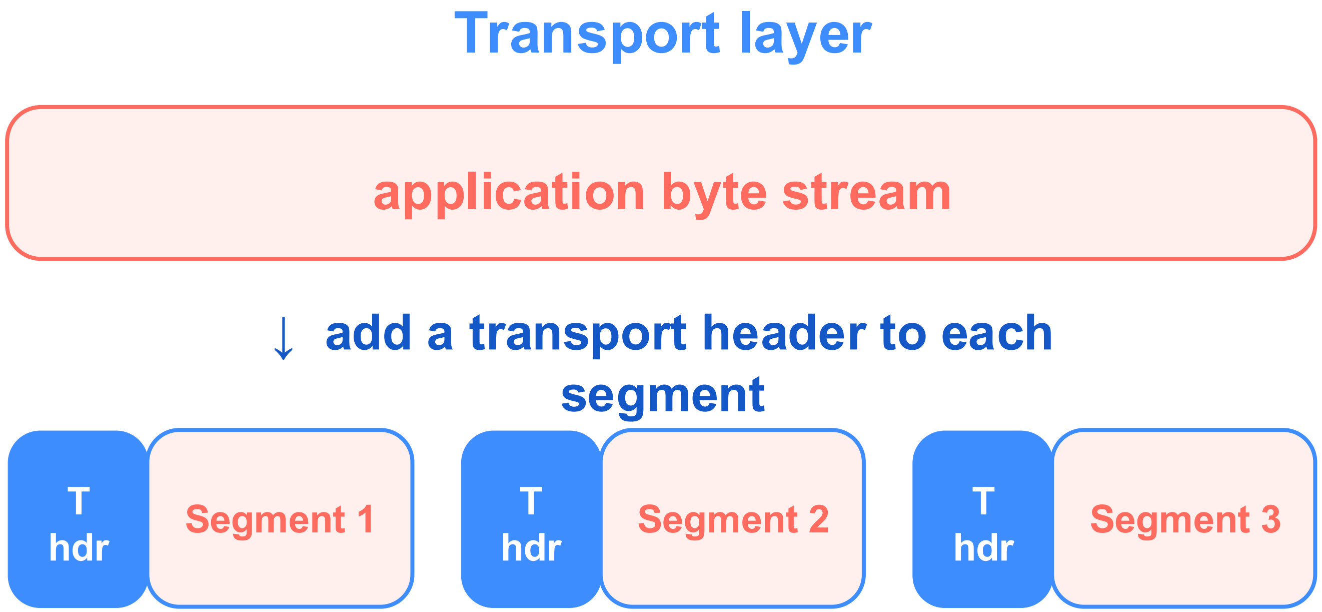
↓ divide into segments

Segment 1 | Segment 2 | Segment 3

2 · It divides the stream into smaller segments.

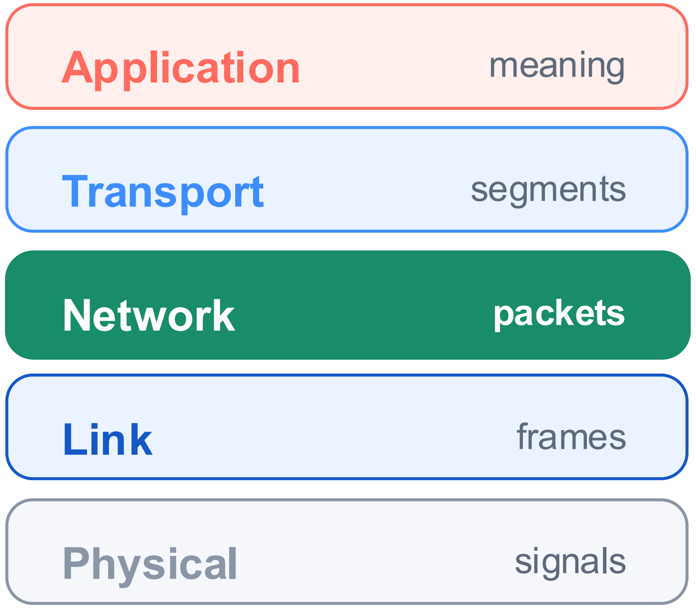
Transport divides data and adds a header to every segment

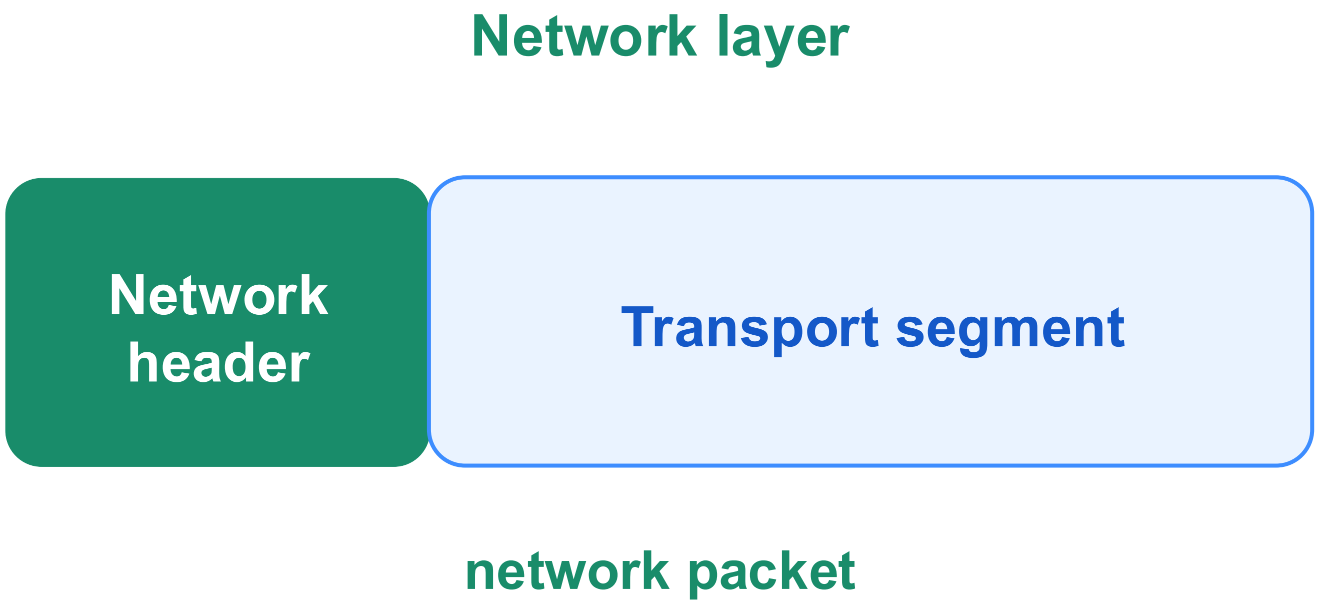




3 · It adds a transport header to every segment.

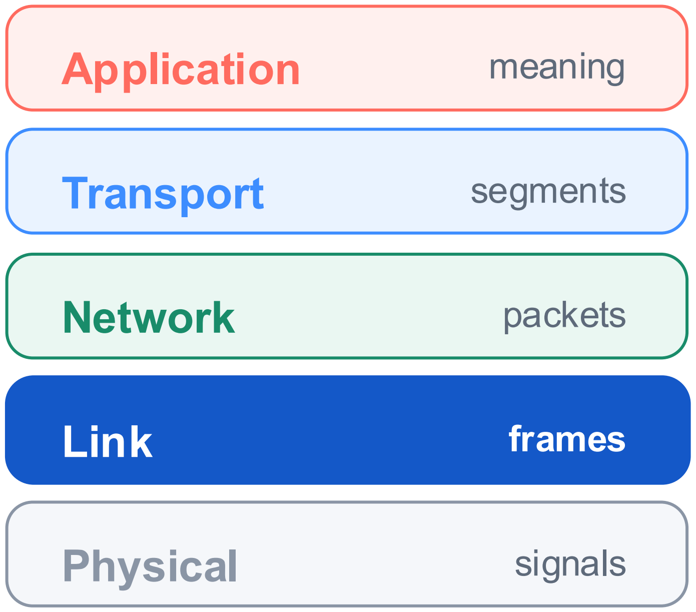
Network wraps a segment in a packet

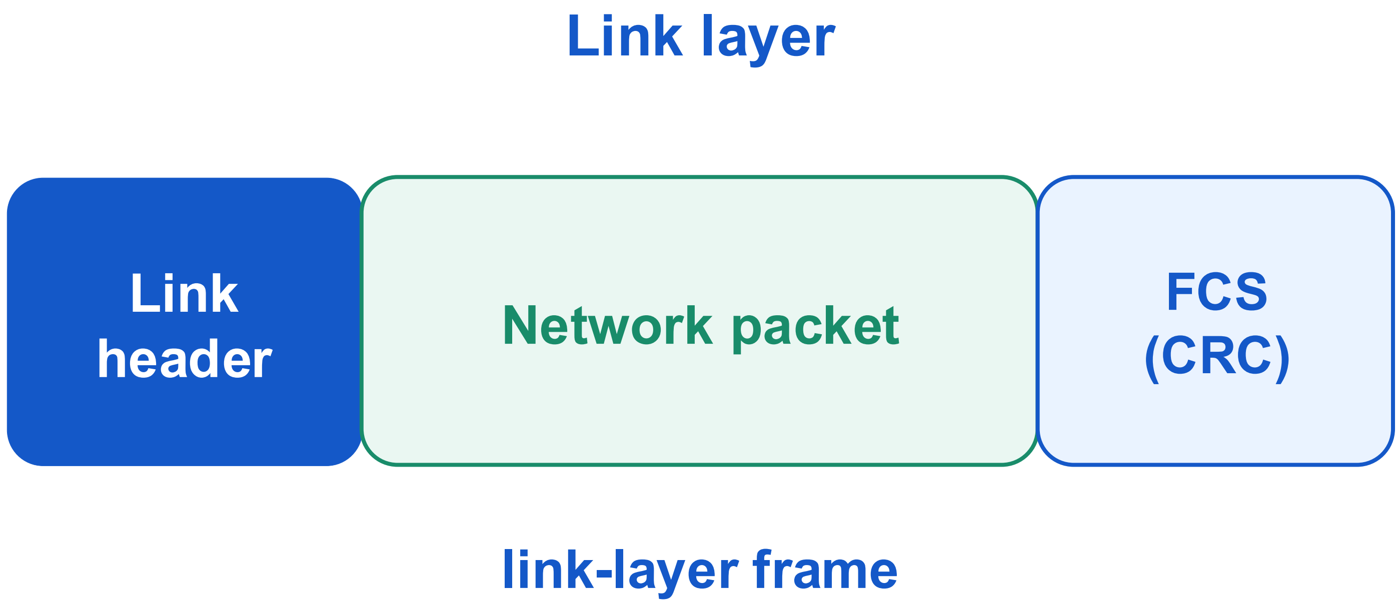




The network header includes the destination IP used for forwarding across networks.

Link wraps a packet in a frame





The frame check sequence (FCS), commonly computed with a CRC, helps detect corrupted frame bits.

Physical converts frame bits into signals

Application	meaning
Transport	segments
Network	packets
Link	frames
Physical	signals

Physical layer

1 0 1 1 0 0 1 0

↓ encode

∿ ∿∿ ∿ ∿∿∿ ∿

electrical · optical · radio

Physical represents frame bits as signals that can cross one real medium.

Encapsulation adds one layer at a time

Application

Data

Transport

Network

Link

1 · The application begins with data and its meaning.

Encapsulation adds one layer at a time

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Application

Data

↓

Transport

T header | Data

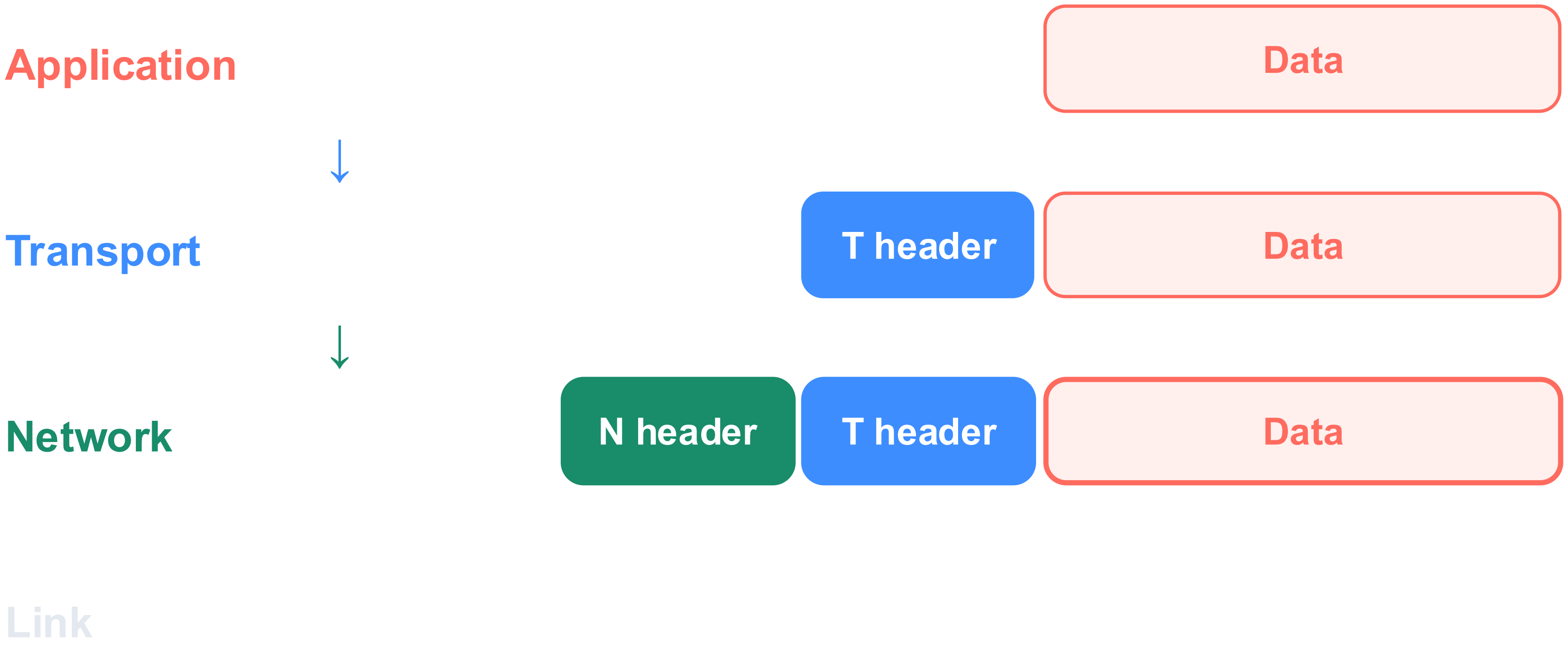
Network

Link

2 · Transport adds a header and creates a segment.

Encapsulation adds one layer at a time

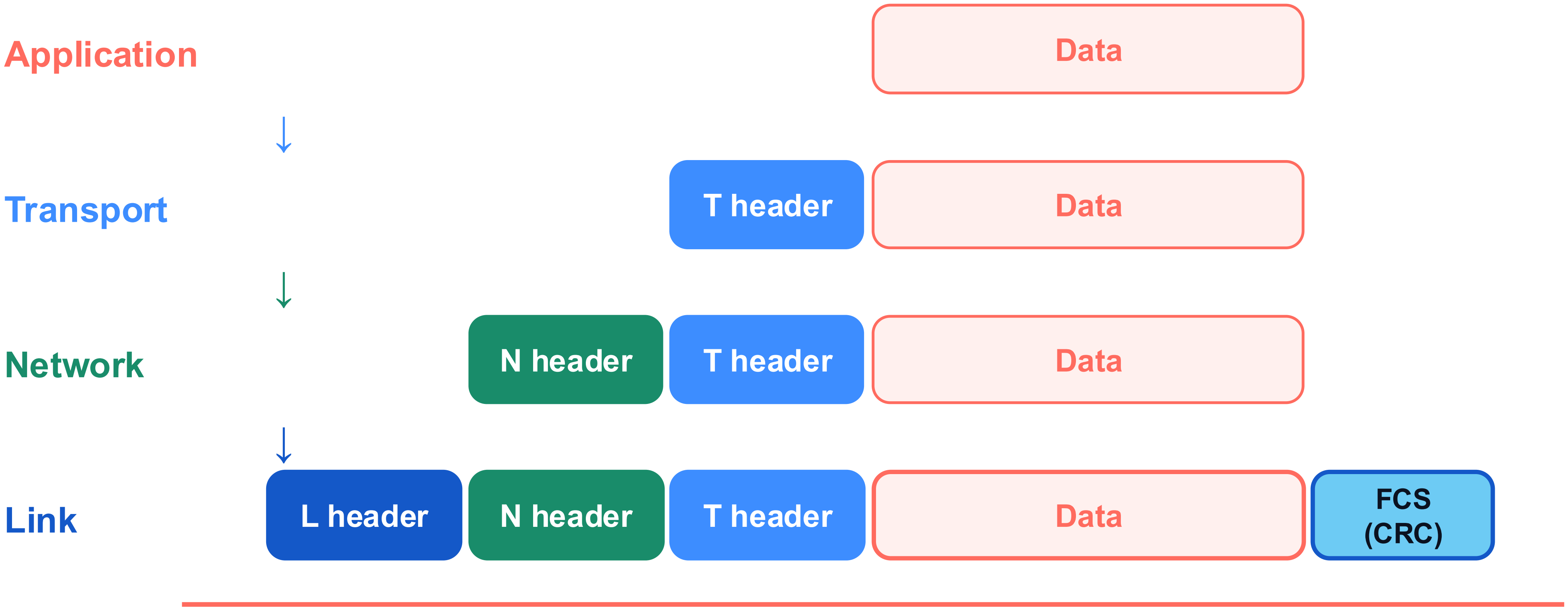
3 / 4



3 · Network adds a header and creates a packet.

Encapsulation adds one layer at a time

4 / 4



4 · Link adds a header and FCS, creating the frame sent on one hop.

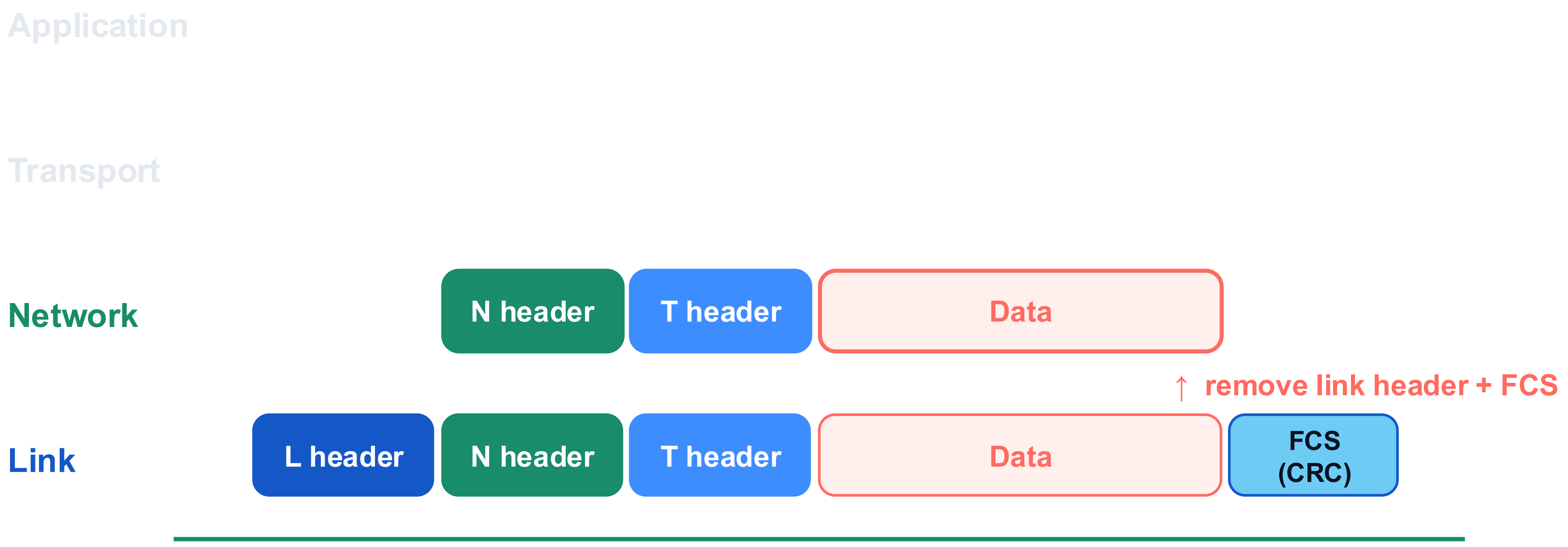
The receiver removes one layer at a time

1 / 4

1 · The receiver begins with the complete link-layer frame.

The receiver removes one layer at a time

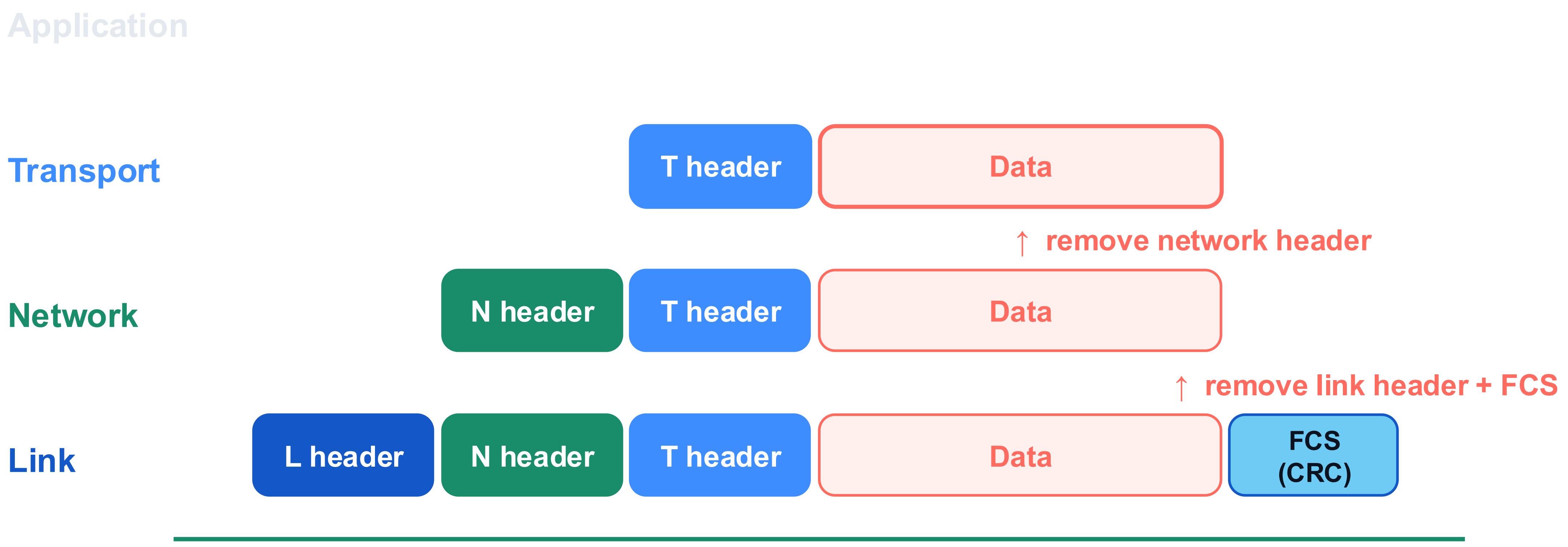
2 / 4



2 · Link verifies the frame, removes link information, and passes up the packet.

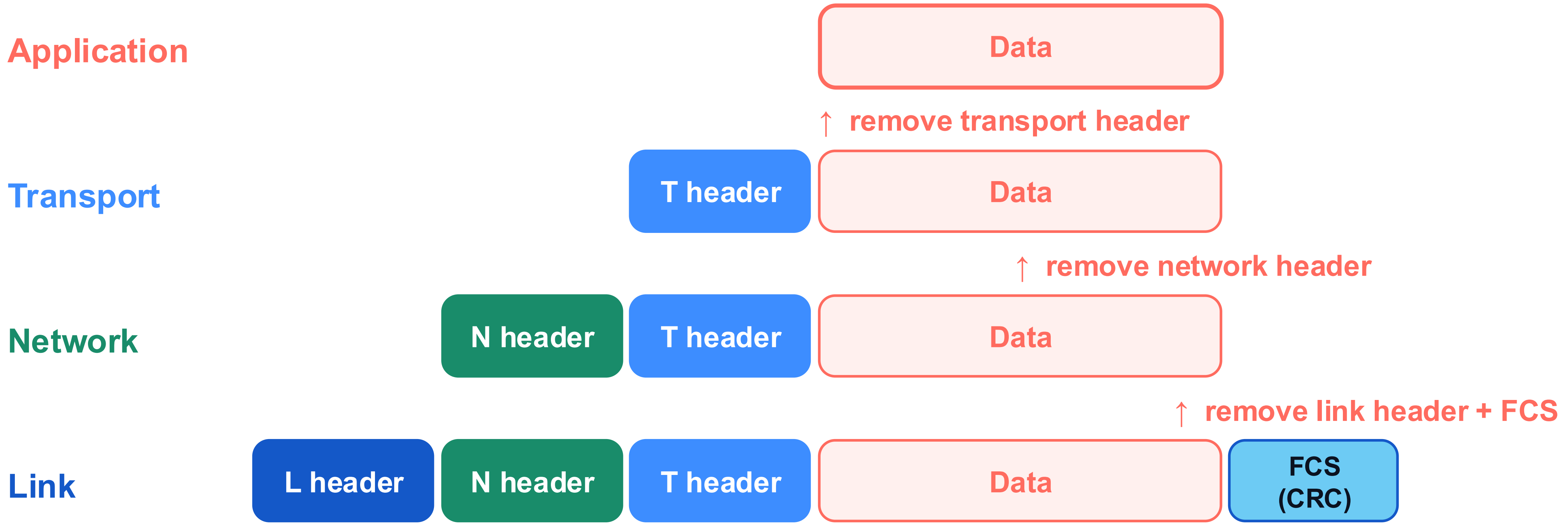
The receiver removes one layer at a time

3 / 4



3 · Network removes its header and passes up the transport segment.

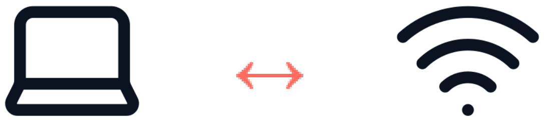
The receiver removes one layer at a time



4 · Transport removes its header and delivers the original data to the application.

Every network depends on real physical links

1 / 3

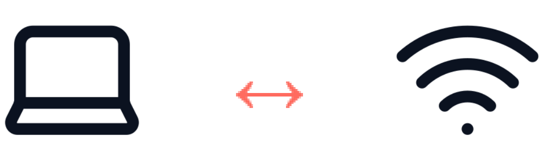


Laptop ↔ access point

radio

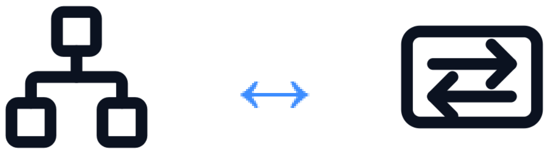
An edge device needs a real radio link to reach its access point.

Every network depends on real physical links



Laptop ↔ access point

radio



Switch ↔ router

copper or fiber

Local network devices also need real wired or optical links between interfaces.

Every network depends on real physical links

Laptop ↔ access point

radio

Switch ↔ router

copper or fiber

Router ↔ router

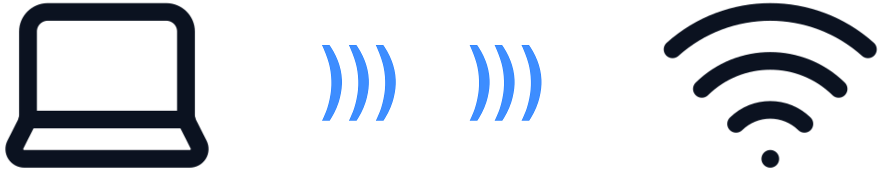
fiber or radio

The core is possible only because routers are joined by physical links.

A physical link connects two adjacent interfaces

A physical link is local: it connects two adjacent network interfaces.

Wi-Fi uses radio for the first physical link

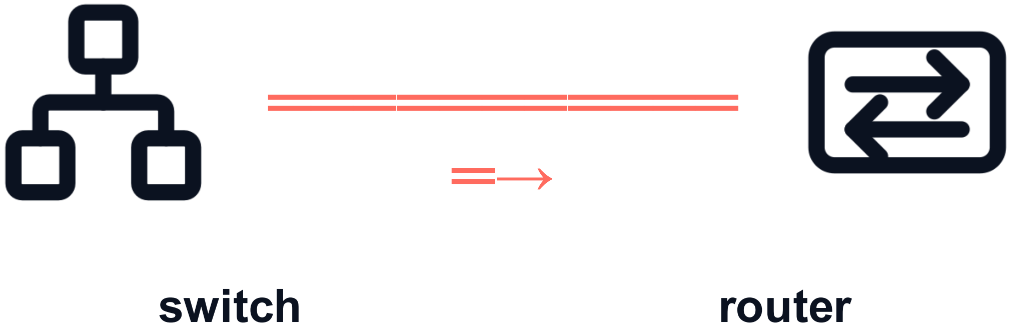


radio link

Laptop: frame bits → radio signal

Access point: radio signal → frame bits

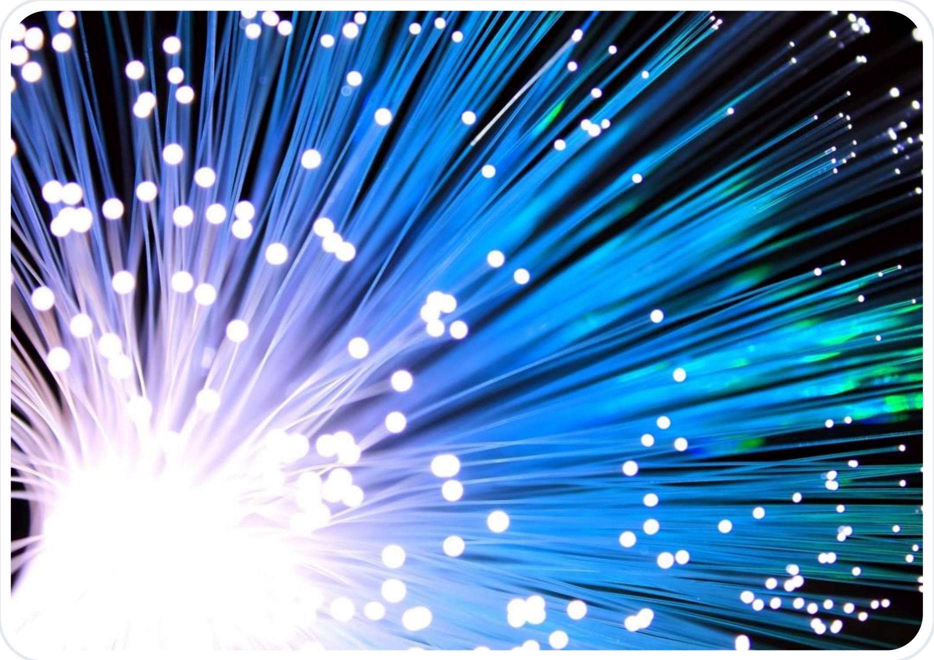
Copper carries electrical signals across a wired link

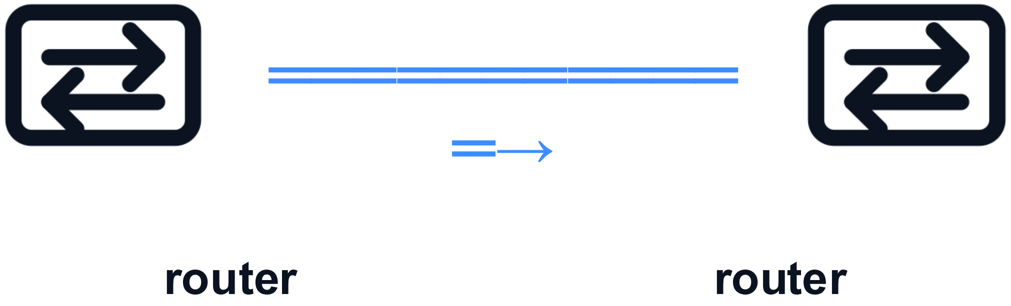


Electrical voltage changes carry bits

Copper is common for short wired Ethernet links inside rooms, labs, and buildings.

Fiber carries light across long, high-capacity links

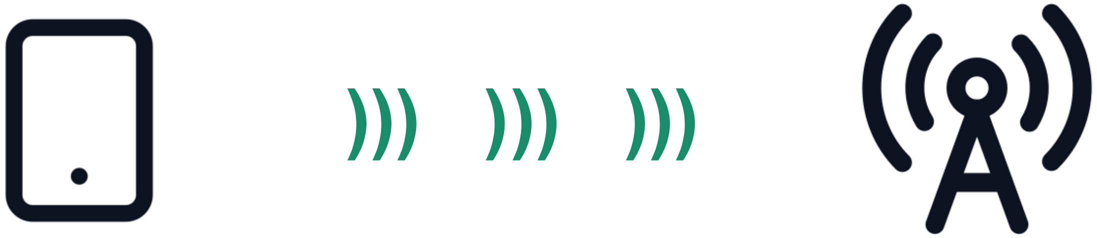




Light pulses carry bits

Fiber supports high capacity and long distances, so it is central to campus backbones and Internet cores.

Radio carries signals without a cable

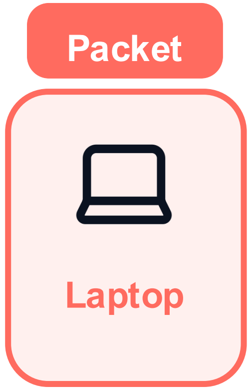


Radio waves carry bits through the air

Wireless links enable mobility, but devices share spectrum and the signal environment changes.

One packet crosses one physical link at a time

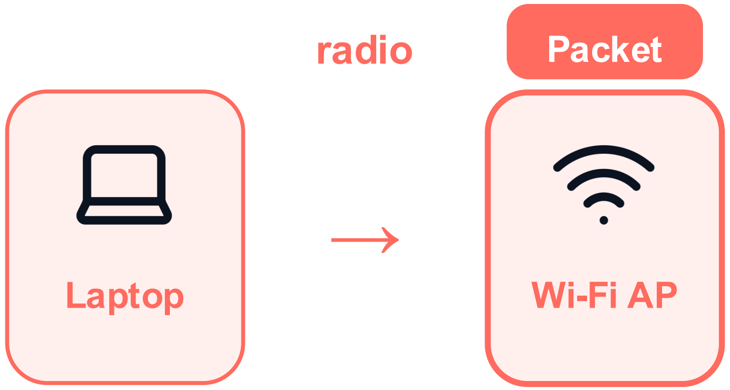
1 / 5



same packet · new physical signal on every link

Now at: Laptop. End-to-end communication is a sequence of local physical links.

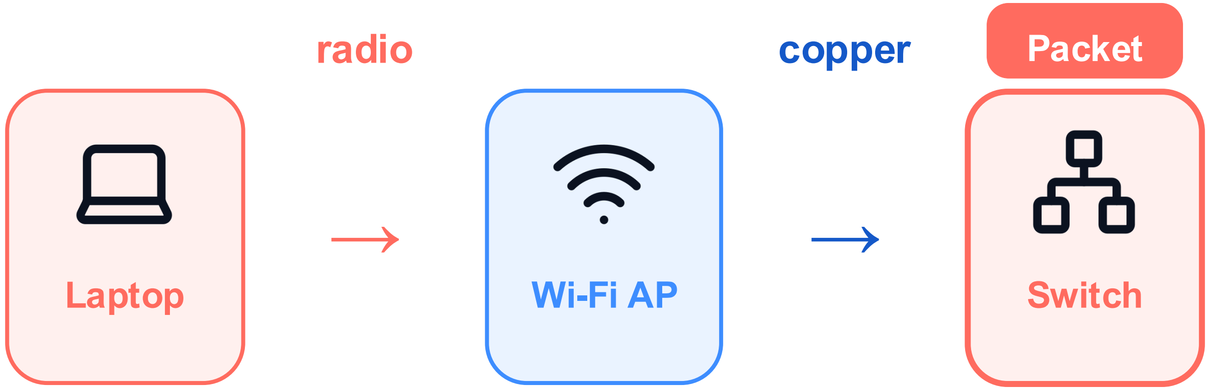
One packet crosses one physical link at a time



same packet · new physical signal on every link

Now at: Wi-Fi access point. End-to-end communication is a sequence of local physical links.

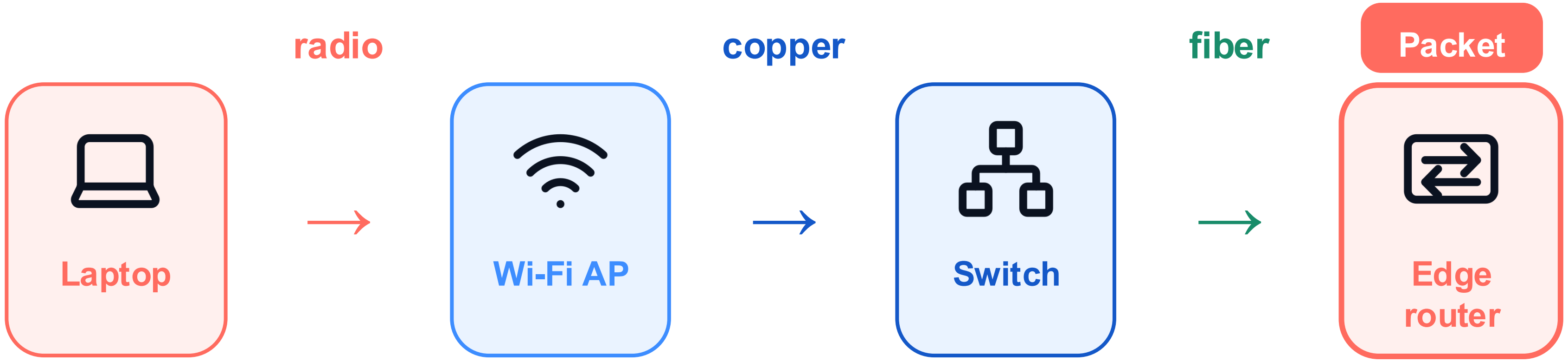
One packet crosses one physical link at a time



same packet · new physical signal on every link

Now at: Campus switch. End-to-end communication is a sequence of local physical links.

One packet crosses one physical link at a time

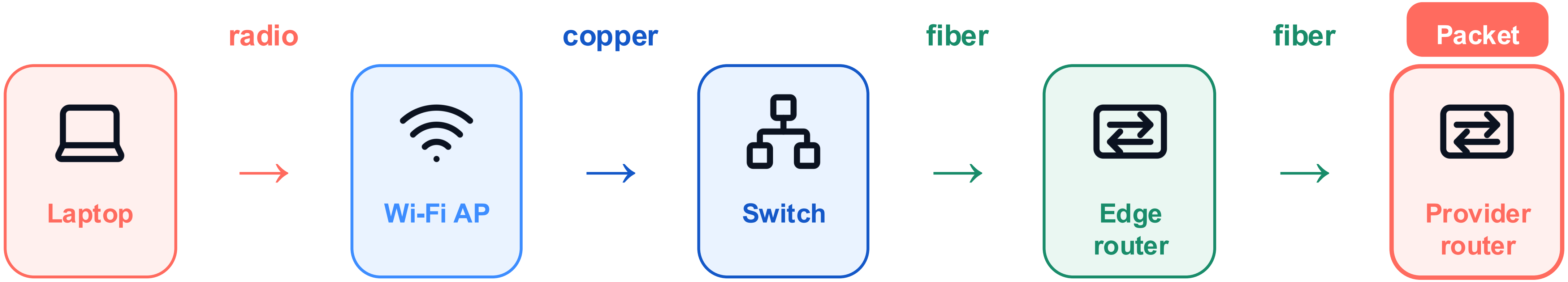


same packet · new physical signal on every link

Now at: Edge router. End-to-end communication is a sequence of local physical links.

One packet crosses one physical link at a time

5 / 5



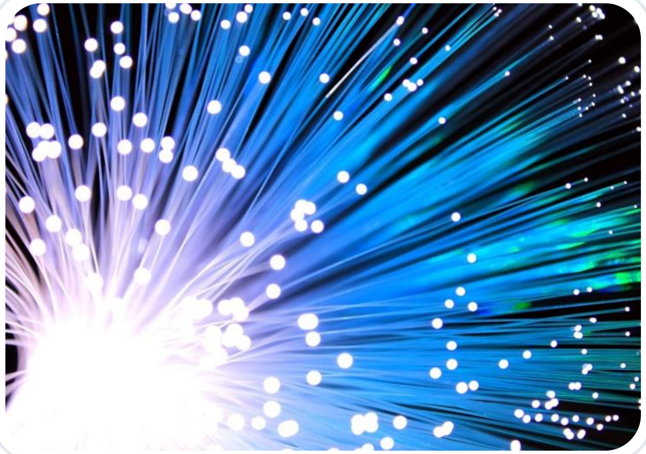
same packet · new physical signal on every link

Now at: Provider router. End-to-end communication is a sequence of local physical links.

Physical media trade off reach, capacity, and mobility

Copper

Electrical signals
Common for short wired links



Fiber

Light pulses
High capacity over long distances

Radio

Electromagnetic waves
Supports wireless access and mobility

Scale and layers describe the same communication

Scale asks where

edge · access · core · remote edge

Layers ask how

Application	meaning
Transport	segments
Network	packets
Link	frames
Physical	signals

One packet journey becomes understandable when we keep both views in sight.

The course moves from applications down to signals

Application	What do network applications say and do?
Transport	What end-to-end service does an application receive?
Network	How are packets forwarded toward a destination?
Link	How does data cross one local hop?
Physical	How are bits carried as electrical, optical, or radio signals?

We start with applications and move downward while keeping the complete path in view.